

Robust Welfare Guarantees in Bilateral Trading Mechanisms*

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Abstract

A bilateral trading mechanism is evaluated according to the level of expected welfare guaranteed when agents play weakly undominated strategies. We show that, if a finite mechanism guarantees expected welfare W , this is also achieved by the truth-telling strategy profile in a revelation mechanism satisfying “local downward incentive compatibility” (LDIC): An agent has no incentive to pretend to be of a slightly “less-eager” (higher-cost or lower-valuation) type. This result yields an upper bound on the welfare guarantee of any finite mechanism. Also, a lower bound is provided by the LDIC revelation mechanism with a monotone trading rule. For a class of the mechanism designer’s priors, the upper bound implies that no mechanism can improve over a posted-price mechanism of Hagerty and Rogerson (1987). For other priors, we provide an example where a monotone LDIC mechanism offers a strictly higher welfare guarantee than any dominant-strategy mechanism.

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1 Introduction

Mechanism design theory examines which social objectives (such as efficiency, fairness, stability, and so on) can be achieved when agents have private information. To predict the possible outcomes of mechanisms, the standard approach is to assume that the agents play a Bayesian-Nash equilibrium (typically with a “common prior”).

This Bayesian-Nash approach is often criticized for the sensitivity of the predicted outcomes of mechanisms to the assumptions about the agents’ beliefs.¹ Namely, this approach relies on the mechanism designer’s knowledge of the agents’ beliefs about each other’s private information, and their (correct) beliefs about each other’s strategies. A mechanism that induces “desirable” outcomes (given any objective of the mechanism designer) in a Bayesian Nash equilibrium may induce undesirable outcomes if the agents have different beliefs about each other’s private information and strategies.²

Given these criticisms, some researchers have investigated more “robust” mechanisms. The standard approach is to consider a mechanism that induces

¹For example, in the context of game theory, Wilson (1987) argues:

Game theory has a great advantage in explicitly analysing the consequences of trading rules that presumably are really common knowledge; it is deficient to the extent it assumes other features to be common knowledge, such as one agent’s probability assessment about another’s preferences or information. [. . .] I foresee the progress of game theory as depending on successive reduction in the base of common knowledge required to conduct useful analyses of practical problems. Only by repeated weakening of common knowledge assumptions will the theory approximate reality.

See also Neeman (2004) and Bergemann and Morris (2005b).

²Cr  mer and McLean (1985) show that the first-best efficiency with full-surplus extraction is possible if there is a commonly known correlated prior over the agents’ valuations (see d’Aspremont, Cremer, and Gerard-Varet (2004) and Kosenok and Severinov (2008) for similar first-best results under budget balance). However, Neeman (2004) argues that this result crucially depends on the “beliefs-determine-preferences” assumption, and Heifetz and Neeman (2006) show that this beliefs-determine-preferences property is “non-generic” in a more general type space (see also Barelli (2009)).

a dominant strategy for each agent. As shown by Ledyard (1979) and Bergemann and Morris (2005b), if there is a unique allocation for each state that the mechanism designer desires to implement, then the mechanism designer must offer a dominant-strategy mechanism.³ This proves to be restrictive, especially in settings that require a balanced budget.⁴

However, if the objective of the mechanism designer is maximization of his expected “utility” (such as expected welfare or profit), the dominant-strategy approach could be too restrictive. In this problem, the mechanism designer does not care which particular outcome is realized, as long as it induces a reasonably high level of his utility. Thus, there may exist a non-dominant-strategy mechanism that “robustly” achieves a strictly higher utility than any dominant-strategy mechanism.

In this paper, we evaluate a mechanism according to a belief-free approach that is similar to the dominant-strategy approach, but we do not require that the agents’ behavior to be uniquely predicted. Specifically, we allow each agent to take any admissible (i.e., not weakly dominated) action. Each agent may have multiple admissible actions in a mechanism, and the mechanism designer’s utility may depend on which of these actions the agents take. A mechanism is said to “guarantee” a level of expected utility W to the mechanism designer if his expected utility is at least W regardless of the admissible actions the agents take in each state. In this sense, we evaluate a mechanism based on the worst-case admissible strategies.

To be specific, in this paper we consider the following bilateral-trading setting. There is a seller who has an object, and a buyer who has money. Both are risk neutral, and hence, first-best efficiency requires that they trade if and only if the buyer’s valuation for the object is higher than the seller’s

³They also show that, in a quasi-linear environment where the mechanism designer only cares about non-monetary allocations, if there is a unique desirable non-monetary allocation rule (e.g., the first-best efficient trading rule), then the mechanism designer must offer a dominant-strategy mechanism (or an ex post incentive compatible mechanism, more generally).

⁴See Laffont and Maskin (1980), for example.

(opportunity) cost. The mechanism designer has a prior as to the economic fundamentals (i.e., the seller's cost and the buyer's valuation for the object), and desires to maximize the expected welfare subject to (ex post) individual rationality and budget balance.

In this setting, Myerson and Satterthwaite (1983) show that first-best efficiency cannot be generally achieved if there is a common independent prior over the agents' private information, and characterize the second-best Bayesian-incentive-compatible mechanism. Hagerty and Rogerson (1987) characterize dominant-strategy mechanisms, and show that a dominant-strategy mechanism essentially reduces to a (randomized) posted-price mechanism: The mechanism designer fixes a trading price at the beginning, each agent simply accepts or rejects this price, and trade occurs if and only if both accept this price. Because the trading price cannot vary with the agents' private information, the expected welfare of the optimal posted-price mechanism is generally much lower than that of the optimal Bayesian-incentive-compatible mechanism.

In this context, we say that a mechanism guarantees expected welfare W if, given whatever admissible actions the agents take under the mechanism, the induced expected welfare is at least W . We study the optimal welfare guarantee achievable by mechanisms with finite message spaces.⁵ The key difficulty in characterizing the optimal welfare guarantee is that a revelation principle does not apply to this problem. That is, an optimal mechanism is not necessarily represented as an incentive-compatible revelation mechanism, and could then be potentially very complicated. To provide an intu-

⁵As shown by Jackson (1992), with infinite message spaces, we can construct a mechanism with an "integer-game" or "tail-chasing" structure, that fully implements essentially any social choice correspondence. In Section 8.1, we construct an infinite mechanism with an integer-game structure that approximates the first-best expected welfare. However, Jackson (1992) argues that such mechanisms have some undesirable properties, and introduce the class of "bounded" mechanisms which do not suffer from these undesirable properties. In this paper, we mainly focus on finite mechanisms, which are necessarily bounded (although there are bounded mechanisms with infinitely many messages).

ition, note that in this problem, the mechanism designer is concerned about the worst-case outcome under a mechanism. Thus (as in the full or weak implementation literature), each type of an agent may have multiple admissible actions in an optimal mechanism, and we cannot simply delete some of them, because this may lead this type to choose an undesirable action that was originally dominated. Thus, an optimal mechanism may necessarily have message spaces that are much larger than the agents' type spaces.

Rather than exactly characterizing the optimal welfare guarantee, this paper derives an upper and a lower bound. These bounds are obtained by revelation mechanisms, although they need not satisfy all of the standard incentive compatibility constraints. These bounds do not generally coincide, but in some cases, they do, and then the optimal mechanism follows.

To obtain the upper bound, we introduce a class of “locally downward incentive compatible” (LDIC) revelation mechanism. In such a mechanism, each agent is asked to report his type (i.e., cost or valuation). Local downward incentive compatibility means that truth-telling weakly dominates pretending to be a “less-eager” (higher-cost or lower-valuation) type that is “adjacent” to his true type.⁶ We show that if a mechanism guarantees expected welfare W , then there is an LDIC revelation mechanism in which the truth-telling strategies attain W . Of course, this revelation mechanism does not necessarily satisfy “global” downward incentive compatibility nor “upward” incentive compatibility. In particular, we do not know what the worst-case admissible strategies are, and so, we do not know the welfare guarantee of this mechanism.

However, if an LDIC revelation mechanism is also monotonic (MLDIC),⁷ then the monotonicity and single-crossing property in the agents' preferences

⁶An “adjacent” type is naturally defined for a discrete type space. With a continuous type space, we show that a finite mechanism partitions each agent's type space into finite intervals, such that any two types in the same interval share the same admissible message. For each type, its adjacent type is defined as a type in the adjacent subinterval.

⁷A revelation mechanism is monotonic if the probability of trading increases as an agent announces more-eager types for any given message of the opponent.

imply that the mechanism also satisfies *global* downward incentive compatibility, i.e., truth-telling weakly dominates pretending to be *any* less-eager type. On the other hand, it does not necessarily satisfy upward incentive compatibility, and thus, an agent may have an incentive to pretend to be a more-eager type. However, because the trading rule is monotonic, any upward deviation can only *increase* the trade, and hence, the expected welfare. Therefore, the welfare guarantee of an MLDIC mechanism is obtained when the agents report their types truthfully (i.e., the worst-case welfare is achieved by the truth-telling strategies). In this sense, the welfare guarantee of an MLDIC mechanism provides a lower bound of the welfare guarantee of the optimal mechanism.

As a special case, a dominant-strategy mechanism is MLDIC, but also needs to satisfy upward incentive compatibility. Thus, our lower bound is (weakly) higher than the welfare guarantee of the optimal dominant-strategy mechanism.

Our results can be summarized as follows:

$$\text{DSIC} \leq \text{MLDIC} \leq \text{Robust welfare guarantee} \leq \text{LDIC},$$

where DSIC is the welfare achieved by the dominant-strategy mechanisms. Recall that the optimal dominant-strategy mechanism is a posted-price mechanism of Hagerty and Rogerson (1987).

Next, we provide a sufficient condition on the distribution over the cost and valuation for which the upper bound implies that no improvement over posted-price mechanisms is possible. This means that, in those cases, there is no (potentially very complicated) mechanism that can guarantee strictly higher expected welfare than the optimal dominant-strategy mechanism. This result provides a foundation for the use of a dominant-strategy mechanism as the optimal robust mechanism.

For other distributions, we demonstrate mechanisms that offer strictly higher welfare guarantees than any dominant-strategy mechanism. For example, we consider a “two-price mechanism”: The seller chooses an allocation between two alternatives, one with a higher trade and a lower price, and the

other with a lower trade and a higher price. The buyer reports the highest acceptable price. Trade occurs at the price the seller has chosen, but only if his price is acceptable to the buyer. In this mechanism, the buyer has a dominant strategy, but the seller does not, and so the expected welfare varies with the seller's choice of admissible actions. Nevertheless, we show that it can guarantee a strictly higher expected welfare than a posted-price mechanism. Also, we show that it is an optimal mechanism if each agent's type space is binary.

The closest precedents in the literature are Jackson (1992) and Börgers (1991). Jackson (1992) derives a necessary condition on (fully) implementable social choice correspondences in admissible strategies, called “strategy resistance”.⁸ In this paper, we derive a stronger necessary condition to obtain the upper bound of the welfare guarantee of any finite mechanism.

Börgers (1991) considers the voting problem of Gibbard (1973) and Satterthwaite (1975), and shows that a non-dictatorial social choice correspondence can be implementable in admissible strategies, but does not consider the problem of maximizing some objective function of the mechanism designer such as expected welfare or profit.

Some recent papers on robust mechanism design consider an “epistemic” approach (for example, see Bergemann and Morris (2005b) and Chung and Ely (2007)). They describe the agents' beliefs about each other explicitly as a part of their “types”, and analyze the Bayesian Nash equilibria for all type spaces. We discuss the relationship between these approaches and ours in Section 8.2 and 8.3. Among others, Chung and Ely (2007) show that, in some auction settings, no mechanism can guarantee a higher expected revenue than the optimal dominant-strategy mechanism. This result is qualitatively similar to our “no-improvement” result over dominant-strategy mechanisms, but the proof does not directly apply to our setting because of the budget balance requirement in bilateral trading.

⁸As discussed, Jackson (1992) considers all bounded mechanisms, which is a larger class of mechanisms than the class of finite mechanisms.

The rest of the paper is structured as follows. In Section 2, we describe the bilateral trading setting, and define the welfare guarantee of a finite mechanism. In Section 3, as an example, we analyze the class of two-price mechanisms. As described above, a two-price mechanism induces a dominant strategy for one of the agents, but not necessarily for the other. We show that, for some distributions of the agents' cost and valuation, there is a two-price mechanism that guarantees strictly higher expected welfare than any dominant-strategy mechanism. In Section 4, we establish an upper and a lower bound for the welfare guarantee of any finite mechanism. In Section 5, we provide a class of distributions for which the upper bound implies that no improvement over the posted-price mechanisms is possible. Section 6 shows that, with binary type spaces, a two-price mechanism is an optimal mechanism, and guarantees strictly higher expected welfare than the optimal posted-price mechanism. In Section 7, we introduce the class of “(fixed) demand-curve” (or “supply-curve”) mechanisms where one of the agents has a dominant strategy, while the other may not, which generalizes two-price mechanisms. We discuss when this class of mechanisms could achieve the lower bound of the optimal welfare guarantee. In Section 8, we discuss how our results could change if: (i) we allow mechanisms with infinite message spaces, (ii) we adopt alternative solution concepts, such as iterative admissibility or an epistemic approach, or (iii) exact budget balance is replaced with “expected” budget balance. Section 9 concludes.

2 Environment

There is a risk-neutral seller ($i = 1$) who has an object, and there is a risk-neutral buyer ($i = 2$) who has money. Let $c \in [0, 1]$ denote the seller's opportunity cost for the object, and let $v \in [0, 1]$ denote the buyer's valuation for the object. We assume that c is the seller's private information, and v is the buyer's private information.

We require that an allocation satisfies ex post individual rationality and

budget balance at each state: An allocation at each state $(c, v) \in [0, 1]^2$ is a pair $(z(c, v), p(c, v))$, where $z(c, v) \in [0, 1]$ is the probability of trading,⁹ and $p(c, v) \in \mathbb{R}$ is the price, or the payment from the buyer to the seller in the event that the buyer obtains the object.¹⁰ The ex post individual rationality requires that (i) no payment is made if there is no trade, and (ii) $p(c, v) \in [c, v]$ if $z(c, v) > 0$. The seller's utility and the buyer's utility at state (c, v) are given by $u_1 = (p(c, v) - c)z(c, v)$ and $u_2 = (v - p(c, v))z(c, v)$, respectively.

Hence, the economic welfare at state (c, v) is given by $z(c, v)(v - c)$, and therefore, the first-best efficiency is achieved when the agents trade if and only if $v \geq c$. We assume that the mechanism designer has a (subjective) prior distribution $F(c, v)$, and desires to maximize the expected welfare:

$$\int_{c,v} z(c, v)(v - c) dF(c, v).$$

2.1 A mechanism and its welfare guarantee

A mechanism is a pair $\langle M, g \rangle$, where $M = M_1 \times M_2$, each M_i is agent i 's message space, which we assume to be finite,¹¹ and $g : M \rightarrow [0, 1] \times \mathbb{R}_+$ is an outcome function, so that $g(m_1, m_2) = (z(m_1, m_2), p(m_1, m_2))$ denotes the allocation (i.e., $z(m_1, m_2)$ denotes the probability of trading and $p(m_1, m_2)$ denotes the price) when agent i reports message $m_i \in M_i$.

⁹Another interpretation is that z represents the time of trading in a continuous-time dynamic bargaining setting. Suppose that the mechanism designer can specify the time of trading τ in a continuous-time model where the agents have the same discount rate r . Then, by setting z so that $z = e^{-r\tau}$, an allocation in this dynamic model is denoted by (z, p) , and therefore, we can effectively design the same mechanism. Copic and Ponsati (2008) provide a similar interpretation of (randomized) posted-price mechanisms of Hagerty and Rogerson (1987) in such a dynamic bargaining setting.

¹⁰Thus, the allocation satisfies budget balance.

¹¹We discuss in Section 8.1 how the results could change with mechanisms with infinitely many messages.

Under an arbitrary mechanism $\langle M, g \rangle$, we allow each agent to take any strategy that is admissible (i.e., not weakly dominated). Let $M_1^{UD}(c) \subseteq M_1$ denote the set of admissible messages for the seller of type c .¹²

$$\begin{aligned} M_1^{UD}(c) = & \{m_1 \in M_1 \mid \nexists m'_1 \in M_1 \text{ s.t.} \\ & \forall m_2 \in M_2, (p(m'_1, m_2) - c)z(m'_1, m_2) \geq (p(m) - c)z(m), \\ & \exists m_2 \in M_2, (p(m'_1, m_2) - c)z(m'_1, m_2) > (p(m) - c)z(m)\}. \end{aligned}$$

Similarly, let $M_2^{UD}(v) \subseteq M_2$ denote the set of messages that are not weakly dominated for the buyer of type v . Define $M^{UD}(c, v) = M_1^{UD}(c) \times M_2^{UD}(v)$.

In the following, we only consider mechanisms that satisfy (ex post) individual rationality. A mechanism is individually rational if for any c, v and $m \in M^{UD}(c, v)$, $p(c, v) \in [c, v]$ if $z(c, v) > 0$. That is, no trade occurs if there is no surplus of trading.

Given a state (c, v) , the probability of trading under the mechanism could vary depending on the agents' message choices. Let $\underline{z}(c, v)$ denote the lowest probability of trading that can be attained among all admissible message profiles for the agents of the types (c, v) . That is,

$$\underline{z}(c, v) = \min_{m \in M^{UD}(c, v)} z(m)(v - c).$$

Definition 1. We say that a mechanism $\langle M, g \rangle$ *guarantees* expected welfare W if^{13,14}

$$W(\Gamma) = \int_{c, v} \underline{z}(c, v)(v - c) dF(c, v) \geq W.$$

¹²Following Börgers (1991) and Jackson (1992), a message is admissible if it is not weakly dominated by any (non-randomized) message even if it is weakly dominated by a randomization of other messages. See Börgers (1993) for a discussion of pure strategy dominance. We discuss later that the results do not essentially change even if we define admissible actions based on mixed strategy dominance.

¹³This condition implicitly requires that $\underline{z}(c, v)(v - c)$ is a measurable function.

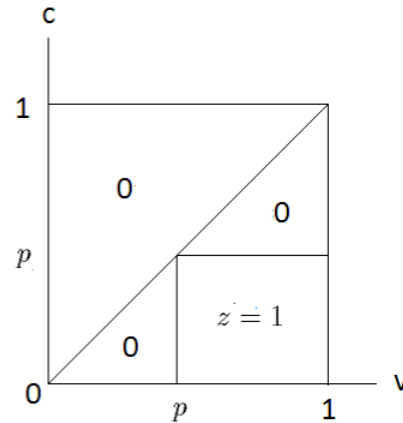
¹⁴We implicitly allow each agent to take his action contingent not only on his own preference type, but also on the opponent's preference type. The idea behind it is that the agents may be able to choose actions contingent on not only their preference types, but also some other (payoff-irrelevant) information that are potentially correlated with the

Recall that $\underline{z}(c, v)$ is the lowest probability of trading that can be induced if the agents choose admissible messages, and thus, $\underline{z}(c, v)(v - c)$ is the lowest possible welfare at (c, v) . Hence the condition means that, whatever admissible messages the agents take, the worst-case expected welfare under the mechanism is at least W . In the following, an “optimal” mechanism refers to a mechanism that attains the highest welfare guarantee.

As a special case, if we consider a dominant-strategy mechanism, an optimal dominant-strategy mechanism is given by a posted-price mechanism (Hagerty and Rogerson (1987)). When p is the price to be posted by the mechanism designer, then a trade occurs if $v > p$ and $c < p$, and a trade does not occur if either $v \leq p$ or $c \geq p$.¹⁵ Thus, the welfare guarantee of a posted-price mechanism is

$$\int_{(c,v) \in [0,p] \times (p,1]} (v - c) dF(c, v).$$

	Reject	Accept
Reject	(0, 0)	(0, 0)
Accept	(0, 0)	(1, p)



opponent’s preference type. Then the agent can effectively choose his action contingent on the opponent’s preference type. Of course, it does not mean that the agents know the opponent’s type.

¹⁵The agents with the threshold types ($c = p$ or $v = p$) have two admissible actions, to accept and reject p . If F is continuous, then these types’ behaviors do not matter, because it is a probability zero event. If F has an atom at these threshold types, the worst-case expected welfare is attained when they reject the price. See Dasgupta, Hammond, and Maskin (1979).

The table above is a posted-price mechanism with price p . If both of the agents accept p , then the trade occurs at price p . Otherwise, no trade. The figure on the right shows the region of trading: $z = 1$ if and only if $c < p$ and $v > p$.

3 Example

In this section, we compare two simple mechanisms. One is a posted-price mechanism, and the other is a “two-price” mechanism, which is not a dominant-strategy mechanism. We show that which one guarantees a higher expected welfare depends on the specific distribution. We provide some intuition for when improvement over a posted-price mechanism would be possible by a non-dominant-strategy mechanism.

We first introduce the following two-price mechanism. The seller can choose either of the two allocations, (z_1, p_1) or (z_2, p_2) , or $(0, 0)$ (no trading) where $z_1 < z_2$, $p_1 > p_2$. This is a slight generalization of a posted-price mechanism, which has only one possible trade allocation. The first allocation has a higher price, but it attains a smaller trade than the second. At the same time, the buyer announces the highest acceptable price $\bar{p} \in \{0, p_1, p_2\}$. $\bar{p} = 0$ corresponds to no-trading. If the seller chooses (z_k, p_k) , and if $\bar{p} \geq p_k$, then the trade occurs at price p_k with probability z_k . No trade occurs if $\bar{p} < p_k$. Thus, the game form is

	$\bar{p} = 0$	$\bar{p} = p_2$	$\bar{p} = p_1$
$(0, 0)$	$(0, 0)$	$(0, 0)$	$(0, 0)$
(z_1, p_1)	$(0, 0)$	$(0, 0)$	(z_1, p_1)
(z_2, p_2)	$(0, 0)$	(z_2, p_2)	(z_2, p_2)

In this mechanism, the buyer can only reject some of the allocations through a report of a threshold price, and therefore, he has a dominant strategy: If $v > p_1$, then he reports p_1 . If $v \in (p_2, p_1)$, then he reports p_2 . If $v < p_2$, then he reports 0.¹⁶

¹⁶If $v = p_1$, then he is indifferent between $\bar{p} = p_1$ and p_2 . If $v = p_2$, then he is indifferent

On the other hand, the seller does not have a dominant strategy, and his best strategy depends on his belief as to the buyer's behavior. First, if he believes that the buyer would choose $\bar{p} = p_2$, then his best action is to choose (i) (z_2, p_2) if $c < p_2$, or (ii) $(0, 0)$ if $c > p_2$.

However, if the seller believes that the buyer would choose $\bar{p} = p_1$ (i.e., if he is "optimistic" about the opponent's behavior), then the seller may attempt to be more greedy. Let c^* be the cost type who is indifferent between (z_1, p_1) and (z_2, p_2) , i.e.,

$$(p_1 - c^*)z_1 = (p_2 - c^*)z_2,$$

or $c^* = \frac{p_2 z_2 - p_1 z_1}{z_2 - z_1}$. Note that $c^* < p_2$. Therefore, the seller's best action is (i) (z_2, p_2) if $c < c^*$, (ii) (z_1, p_1) if $c \in (c^*, p_1)$, or (iii) $(0, 0)$ if $c > p_1$.

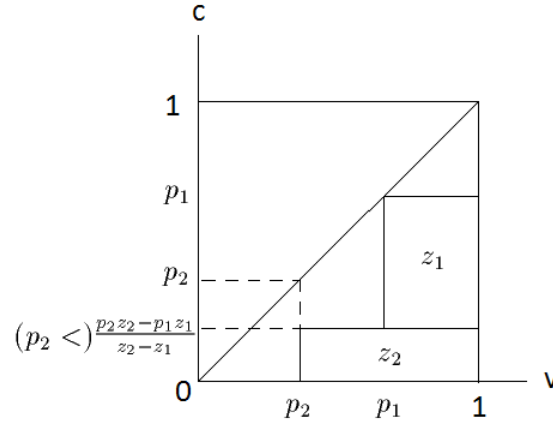
The worst-case expected welfare is attained when the seller is "optimistic" about the buyer's behavior, and attempts to be greedy. Given any $F(c, v)$, the welfare guarantee of the mechanism is given by

$$\int_{c=0}^{c^*} \int_{v=p_2}^1 z_2(v - c) dF(c, v) + \int_{c=c^*}^{p_1} \int_{v=p_1}^1 z_1(v - c) dF(c, v).$$

between $\bar{p} = 0$ and p_2 .

	Opt-out	p_2	p_1	
Opt-out	(0, 0)	(0, 0)	(0, 0)	← Admissible for $c > p_1$
(z_1, p_1)	(0, 0)	(0, 0)	(z_1, p_1)	← Admissible for $c \in (p_1, \frac{p_2 z_2 - p_1 z_1}{z_2 - z_1})$
(z_2, p_2)	(0, 0)	(z_2, p_2)	(z_2, p_2)	← Admissible for $c \in (0, p_2)$

The seller has two admissible messages if $c \in (\frac{p_2 z_2 - p_1 z_1}{z_2 - z_1}, p_2)$



The figure shows the worst-case scenario under the two-price mechanism. Trade occurs with probability z_1 if $c \in (c^*, p_1)$ and $v > p_1$, and with probability z_2 if $c < c^*$ and $v > p_2$.

In the following, we compare this two-price mechanism and the posted-price mechanism under different distributions.

3.1 Uniform distribution

First, let v be uniformly distributed on $[\frac{1}{4}, 1]$, and independently, c be uniformly distributed on $[0, \frac{3}{4}]$.

The optimal posted-price mechanism sets $p = \frac{1}{2}$, and yields the welfare guarantee

$$\int_{c=0}^{\frac{1}{2}} \int_{v=\frac{1}{2}}^1 (v - c) \left(\frac{4}{3}\right)^2 dv dc = \frac{2}{9},$$

which is roughly 77% of the first-best welfare.¹⁷

Under the two-price mechanism, the welfare guarantee is

$$\int_{c=0}^{c^*} \int_{v=p_2}^1 z_2(v-c) \left(\frac{4}{3}\right)^2 dvdc + \int_{c=c^*}^{p_1} \int_{v=p_1}^1 z_1(v-c) \left(\frac{4}{3}\right)^2 dvdc,$$

where $c^* = \frac{p_2 z_2 - p_1 z_1}{z_2 - z_1}$, and it is maximized if $z_1 = p_1 = 0$, $z_2 = 1$, and $p_2 = \frac{1}{2}$. However, this is equivalent to the optimal posted-price mechanism. Therefore, under this uniform distribution, a two-price mechanism cannot be better than a posted-price mechanism.

3.2 “Almost two-type” distribution

Here, we consider a different situation, where the types are concentrated on the extreme points. That is, $\Pr(c = 0) = \Pr(c = \frac{3}{4}) = \frac{1-\varepsilon}{2}$, $\Pr(v = \frac{1}{4}) = \Pr(v = 1) = \frac{1-\varepsilon}{2}$, and $(c, v) \in (0, \frac{3}{4}) \times (\frac{1}{4}, 1)$ with density $\frac{4}{3}\varepsilon$, where $\varepsilon > 0$ is small.¹⁸

We first obtain the optimal posted-price mechanism. Observe that the optimal price is either $\frac{1}{4}$ or $\frac{3}{4}$. If $p = \frac{1}{4}$, then the expected welfare is, if ε is close to zero, approximately

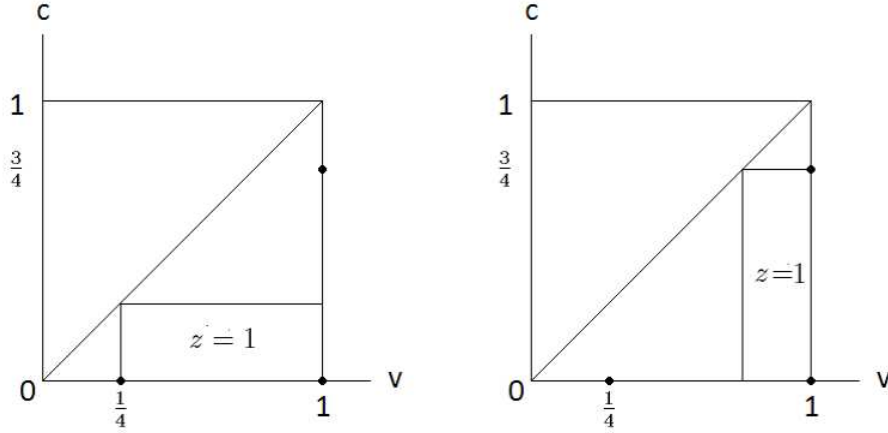
$$\frac{1}{4} \cdot \frac{1}{4} + \frac{1}{4} = \frac{5}{16},$$

and the same welfare obtains if $p = \frac{3}{4}$.¹⁹

¹⁷The first-best welfare is $\frac{31}{108}$.

¹⁸It is just for simplicity that we assume a distribution with an atom. A similar analysis applies to an atomless distribution.

¹⁹More rigorously, $\frac{5}{16}$ is the limit of the welfare guarantees of posted-price mechanisms with $p \uparrow \frac{1}{4}$ (or $p \downarrow \frac{3}{4}$). In this subsection we informally say that the optimal mechanism sets $p = \frac{1}{4}$ (or $\frac{3}{4}$) if such a sequence of mechanisms clearly exists. We will be more rigorous in the other sections.



Now we consider the two-price mechanism. As we will see later, the optimal two-price mechanism sets $(z_1, p_1) = (\frac{1}{3}, \frac{3}{4})$, and $(z_2, p_2) = (1, \frac{1}{4})$. In this mechanism, the buyer has a dominant strategy, but the seller does not. For the seller with $c > \frac{1}{4}$, the only admissible action is to choose (z_1, p_1) . For $c \in (\frac{1}{4}, 1)$, both (z_1, p_1) and (z_2, p_2) are admissible: If he believes the high price p_1 is acceptable for the buyer, then he would choose (z_1, p_1) , and if he believes only the low price p_2 is acceptable for the buyer, then (z_2, p_2) . If $c = 0$, then (z_2, p_2) is the only admissible action.

The worst-case expected welfare occurs when the seller with any positive cost chooses (z_1, p_1) , and then we obtain, if ε is close to zero,

$$\frac{1}{3} \cdot \frac{1}{4} \cdot \frac{1}{4} + \frac{1}{4} \cdot \frac{1}{4} + \frac{1}{4} = \frac{1}{3},$$

which is greater than $\frac{5}{16}$. Therefore, the two-price mechanism attains improvement over the optimal posted-price mechanism.

This example yields some interesting lessons. First, there can be a (relatively simple) non-dominant-strategy mechanism that guarantees a higher expected welfare than a dominant-strategy mechanism. Second, such a mechanism may exist when the prior $F(c, v)$ assigns relatively larger weight on “less eager” types (higher-cost sellers or lower-valuation buyers). In a posted-price mechanism, these type may not trade at all, while in the two-price mechanism, they could (for example, the seller with $c = \frac{3}{4}$ or the buyer with $v = \frac{1}{4}$).

One question is whether there could be more complicated mechanisms that guarantee even higher expected welfare than the posted-price mechanism or the two-price mechanism analyzed here. Or, for general priors, which mechanism is optimal among all mechanisms. We partially answer these questions by establishing an upper and a lower bound of the optimal welfare guarantee. When applied to the examples, for the first one with a uniform distribution, our results imply that no mechanism (no matter how complicated) can improve over the optimal posted-price mechanism. For the second example with an “almost two-type” distribution, we show that the two-price mechanism obtained here is nearly optimal so that no mechanism (no matter how complicated) can improve by more than ε than the two-price mechanism.

It is also interesting to compare the welfare guarantees in the two examples above with the expected welfare under the optimal Bayesian mechanism of Myerson and Satterthwaite (1983). In the uniform case, the optimal Bayesian mechanism attains $\frac{1}{4}$, which is 87% of the first-best welfare,²⁰ while the optimal welfare guarantee is $\frac{2}{9}$, which is 77% of the first-best welfare. This difference can be interpreted as the “price of robustness”.²¹

In the second case, the optimal Bayesian mechanism achieves nearly the

²⁰Specifically, we set the probability of trading is $z(c, v) = 1$ if $v > c + \frac{1}{4}$ and zero otherwise, and the price is $p(c, v) = \frac{c+v}{3} + \frac{1}{6}$. This is derived by maximizing the weighted virtual surplus as in an example in Myerson and Satterthwaite (1983). See also Chatterjee and Samuelson (1983).

²¹Moreover, if c and v are correlated, then the optimal Bayesian mechanism may be able to achieve the first-best efficiency. Kosenok and Severinov (2008) construct such a mechanism when the agents’ type spaces are finite.

first-best welfare, which is approximately $\frac{3}{8}$ if ε is small.²² The optimal posted-price mechanism guarantees $\frac{5}{16}$, 83% of the first-best welfare, while the optimal two-price mechanism guarantees $\frac{1}{3}$, 89% of the first-best welfare. The difference between these two values quantifies the welfare loss due to the restriction to dominant-strategy mechanisms: The “price of robustness” is 11% among all mechanisms, while it is 17% if we are restricted to dominant-strategy mechanisms. The following table summarizes the results. The percentage in each cell is relative to the first-best expected welfare.

	Example 1 (Uniform)	Example 2 (Almost two types)
Bayesian-Nash	87%	100%
Robust welfare guarantee	77%	89%
Dominant-strategy	77%	83%

4 Two bounds for the optimal welfare guarantee

A key difficulty in characterizing the optimal welfare guarantee is that a revelation principle does not apply to this problem. That is, an optimal mechanism is not necessarily represented as a revelation mechanism.²³

²²Consider the case with $\varepsilon = 0$. Then the first-best welfare is achieved by the following (revelation) mechanism:

	$v = \frac{1}{4}$	$v = 1$
$c = \frac{3}{4}$	$(0, 0)$	$(1, \frac{3}{4})$
$c = 0$	$(1, \frac{1}{4})$	$(1, \frac{1}{2})$

The first-best welfare is $\frac{3}{8}$. If $\varepsilon > 0$, then the exact first-best is impossible, but the expected welfare can be close to the first-best level.

²³In this problem, the mechanism designer is concerned about the worst-case expected welfare, and therefore, as in full or weak implementation literature, each type of an agent may have multiple admissible actions in an optimal mechanism, even if most of them are “irrelevant” to calculate the worst-case expected welfare. However, we cannot simply

In this paper, instead of directly characterizing the optimal welfare guarantee, we use revelation mechanisms to establish upper and lower bounds.

We first introduce a revelation mechanism with “threshold types”.

Definition 2. A mechanism $\Gamma^R = \langle M, g \rangle$ is a *revelation mechanism with thresholds* $\{c_1, \dots, c_J\}$ and $\{v_1, \dots, v_K\}$ if (i) $0 \leq c_1 < \dots < c_J \leq 1 (\equiv c_{J+1})$, $0 (\equiv v_0) \leq v_1 < \dots < v_K \leq 1$, (ii) $M_1 = M_2 = [0, 1]$, and (iii) there is a vector of allocations $(z_{jk}, p_{jk})_{j=1, \dots, J+1; k=0, \dots, K}$ such that

$$g(c, v) = (z_{jk}, p_{jk}), \text{ if } c \in C_j, v \in V_k,$$

where $C_1 = [0, c_1]$, $C_j = (c_{j-1}, c_j]$ for $j = 2, \dots, J+1$, $V_k = [v_k, v_{k+1})$ for $k = 0, \dots, K-1$, and $V_K = [v_K, 1]$.²⁴

That is, under the revelation mechanism with threshold types, each agent is asked to report a type, but the allocation does not change as long as his type is in the same interval, $c \in C_j$ and $v \in V_k$. This restriction is because we are interested in finite mechanisms.

Suppose that the agents report their types truthfully in this mechanism.²⁵ Then, the following expected welfare is obtained.

$$\begin{aligned} \overline{W}(\Gamma^R) &= \int_{c,v} (v - c)z(c, v) dF(c, v) \\ &= \sum_{j=1}^{J+1} \sum_{k=0}^K \int_{(c,v) \in C_j \times V_k} (v - c)z_{jk} dF(c, v). \end{aligned}$$

We call $\overline{W}(\Gamma^R)$ the *truth-telling expected welfare* of Γ^R .

Next, we define the “local downward incentive compatibility” of the revelation mechanism with thresholds.

delete them, because deleting them might lead this type to take an undesirable action which was originally dominated. In this sense, an optimal mechanism may necessarily have much larger message spaces than the agents’ type spaces.

²⁴Note that $\{C_j\}_{j=1}^{J+1}$ and $\{V_k\}_{k=0}^K$ are partitions of $[0, 1]$.

²⁵We do not claim that such a behavior is incentive compatible. Indeed, we discuss later that it is not necessarily the case.

Definition 3. A revelation mechanism with thresholds $\{c_1, \dots, c_J\}$ and $\{v_1, \dots, v_K\}$ is locally downward incentive compatible (LDIC) if the following conditions are satisfied: For each $j = 1, \dots, J$ and $k = 1, \dots, K$,

$$\begin{aligned}(p_{jk} - c_j)z_{jk} &\geq (p_{j+1,k} - c_j)z_{j+1,k}, \\ (v_k - p_{jk})z_{jk} &\geq (v_k - p_{j,k-1})z_{j,k-1}, \\ (p_{J+1,k} - 1)z_{J+1,k} &= p_{j0}z_{j0} = 0.\end{aligned}$$

The first inequality means that the seller of each threshold type c_j weakly prefers (z_{jk}, p_{jk}) to $(z_{j+1,k}, p_{j+1,k})$ for any k . Thus, the truth-telling of his type either (i) weakly dominates pretending to be a slightly less eager (i.e., higher-cost) type, or (ii) he is indifferent between the two.²⁶ Similarly, the second inequality means that the buyer of each threshold type v_k weakly prefers (z_{jk}, p_{jk}) to $(z_{j,k-1}, p_{j,k-1})$ for any j . The last equations mean that the utilities of the least-eager types are always zero.

We now show that for any mechanism $\langle M, g \rangle$, there is an LDIC revelation mechanism with some threshold types such that the agents' truth-telling strategy profile under the revelation mechanism provides an upper bound of the welfare guarantee of $\langle M, g \rangle$.

Let $S_1 \times S_2 \subseteq [0, 1]^2$ denote the support of F .

Theorem 1. For any mechanism $\Gamma = \langle M, g \rangle$, there is an LDIC revelation mechanism Γ^R with thresholds $\{c_1, \dots, c_J\}$ and $\{v_1, \dots, v_K\}$ such that $c_j \in S_1$ for each j , $v_k \in S_2$ for each k ,²⁷ and $\overline{W}(\Gamma^R) \geq W(\Gamma)$.

The idea of the proof when $S_1 = S_2 = [0, 1]$ is the following. See the appendix for the proof.

First, we can show that a finite mechanism $\Gamma = \langle M, g \rangle$ partitions the buyer's type space $[0, 1]$ into finite subintervals $\tilde{V}_0, \dots, \tilde{V}_K$ so that for each $k = 0, \dots, K$,²⁸ any two types $v, v' \in \tilde{V}_k$ are "strategically equivalent" in

²⁶For non-threshold types, they have no incentive to locally deviate, because any such deviation does not change the allocation.

²⁷If F has a finite support, then these conditions mean that the number of the threshold types for each agent is at most the number of the agent's types supported in F .

²⁸Without loss of generality, we assume $\tilde{V}_k < \tilde{V}_{k+1}$ in the strong set order.

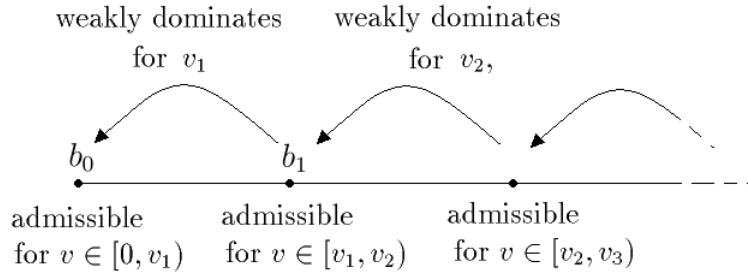
the sense that $M_2^{UD}(v) = M_2^{UD}(v')$.²⁹ Let $\tilde{v}_k = \inf \tilde{V}_k$ be the threshold type between $\tilde{V}_{k-1}$ and $\tilde{V}_k$ for each k . By construction, $\tilde{v}_0 = 0$.

Then, we select messages in M_2 as follows.

(Step 0) Take any message $b_0 \in M_2$ that is admissible for $v \in \tilde{V}_0$. We can choose b_0 so that b_0 is also admissible for $\tilde{v}_0 = 0$.

(Step $t = 1, \dots, K$) Take $b_t \in M_2$ as follows: (i) If b_{t-1} is admissible for $v \in \tilde{V}_t$, then let $b_t = b_{t-1}$. (ii) If b_{t-1} is not admissible, then there must be $b_t \in M_2$ that weakly dominates b_{t-1} and is admissible for $v \in \tilde{V}_t$. Also, we can choose b_t so that b_t is also admissible for $\tilde{v}_t$. For $\tilde{v}_t$, either b_t weakly dominates b_{t-1} , or he is indifferent between b_{t-1} and b_t for any $a \in M_1$.³⁰

Note that this procedure induces a finite sequence of messages $b_0, \dots, b_K$. Define $V_k = [\tilde{v}_k, \tilde{v}_{k+1})$ for $k \leq K-1$ and $V_K = [v_K, 1]$. Then, by construction, b_k is admissible for $v \in V_k$ for each k .



For the seller, we apply the same procedure, except that we start from the highest cost type $c = 1$ and step down. We obtain a finite subintervals $C_1, \dots, C_J$ so that $C_1 = [0, c_1]$ and $C_j = (c_{j-1}, c_j]$ for $j \geq 2$, a sequence of threshold types $\tilde{c}_1, \dots, \tilde{c}_J$, and a finite sequence of messages $a_1, \dots, a_{J+1}$ such

²⁹We can always find such a finite partition, because the set of types of the buyer for whom $b \in M_2$ is admissible is given by

$$\bigcap_{b' \neq b} \left[\left(\bigcup_{a \in M_1} \{v | (v - p(a, b))z(a, b) > (v - p(a, b'))z(a, b')\} \right) \cup \left(\bigcap_{a \in M_1} \{v | (v - p(a, b))z(a, b) = (v - p(a, b'))z(a, b')\} \right) \right],$$

which consists of finitely many subintervals of $[0, 1]$.

³⁰For example, in a posted-price mechanism with price p , the threshold type $v = p$ is indifferent between accepting and rejecting the price.

that (i) a_j is admissible for $c \in C_j$, and (ii) for $\tilde{c}_j$, a_j weakly dominates a_{j+1} , or he is indifferent between a_j and a_{j+1} .

Let $(z_{jk}, p_{jk}) = g(a_j, b_k)$ for each $(c, v) \in C_j \times V_k$.

Now, we define a revelation mechanism Γ^R with threshold types $\tilde{c}_1, \dots, \tilde{c}_J$ and $\tilde{v}_1, \dots, \tilde{v}_K$ so that, for each $(c, v) \in C_j \times V_k$, $(z(c, v), p(c, v)) = (z_{jk}, p_{jk})$. Then, by construction, it is LDIC: For each j, k ,

$$\begin{aligned}(p_{jk} - c_j)z_{jk} &\geq (p_{j+1,k} - c_j)z_{j+1,k}, \\ (v_k - p_{jk})z_{jk} &\geq (v_k - p_{j,k-1})z_{j,k-1}, \\ (p_{J+1,k} - 1)z_{J+1,k} &= p_{j0}z_{j0} = 0.\end{aligned}$$

Also, its truth-telling welfare is

$$\begin{aligned}\overline{W}(\Gamma^R) &= \int_{c,v} (v - c)z(c, v) dF(c, v) \\ &= \sum_{j=1}^{J+1} \sum_{k=0}^K \int_{(c,v) \in C_j \times V_k} (v - c)z_{jk} dF(c, v) \\ &\geq W(\Gamma),\end{aligned}$$

where the inequality comes because a_j is admissible for $c \in C_j$ and b_k is admissible for $v \in V_k$ in Γ .

Remark 1. When $S_i \subsetneq [0, 1]$, then we need to modify the proof slightly so that each threshold is in S_i . Specifically, if the above mentioned procedure induces a threshold type $\tilde{v}_k$ that is not in S_2 , then we replace this threshold with $\tilde{v}'_k = \min\{v \in S_2 | v \geq \tilde{v}_k\}$. Similarly, if the above mentioned procedure induces a threshold type $\tilde{c}_j$ that is not in S_1 , then we replace this threshold with $\tilde{c}'_j = \max\{c \in S_1 | c \geq \tilde{c}_j\}$. See the appendix for the proof.

Remark 2. In the full implementation context in admissible strategies, Jackson (1992) derives a necessary condition on implementable social choice correspondences, called “strategy-resistance”. To obtain the condition, he uses a similar argument, but considers any pair of types and admissible messages, rather than a sequence of admissible messages. Here, taking a sequence of

admissible messages for each agent, we obtain a stronger implication than Jackson (1992).

Note that we do not claim that this LDIC revelation mechanism actually guarantees its truth-telling expected welfare $\overline{W}(\Gamma^R)$. First, the LDIC condition is only a local condition, not global. Therefore, for the seller of $c \in (c_{j-1}, c_j)$, for example, it may be admissible to report $c \in (c_{j-3}, c_{j-2})$, or lower. Second, the LDIC condition is only downward, not upward. Therefore, even locally, an agent may have an incentive to report a higher type than his true type. In this sense, the welfare guarantee obtained in Theorem 1 is only an upper bound.

However, the LDIC revelation mechanism actually guarantees $\overline{W}(\Gamma^R)$ in the following case.

Theorem 2. Suppose that F is absolutely continuous. If an LDIC revelation mechanism Γ^R with some thresholds is monotonic, i.e., z_{jk} is non-increasing in j and non-decreasing in k , then we have $W(\Gamma^R) = \overline{W}(\Gamma^R)$. That is, the welfare guarantee is obtained by the truth-telling strategy profile.

The idea of the proof is simple. Suppose that Γ^R has a monotone trading rule. Then, because the agents' preferences satisfy the single-crossing property, the LDIC condition also implies the global downward incentive compatibility. That is, truth-telling weakly dominates pretending to be *any* less-eager (higher-cost or lower-valuation) type. The upward incentive compatibility may not be satisfied, i.e., an agent of some type may want to pretend to be a more-eager (lower-cost or higher-valuation) type. However, if the allocation is monotonic, such a deviation only *increases* the probability of trading, which is good in terms of the welfare (as long as it does not violate the individual rationality). Therefore, in such a mechanism, the welfare guarantee of a direct revelation mechanism is achieved by the truth-telling strategy profile.

The only subtlety is when all the LDIC conditions for some threshold

type, say c_j , are satisfied with equality. That is,

$$(p_{jk} - c_j)z_{jk} = (p_{j+1,k} - c_j)z_{j+1,k},$$

for all k . Then, this type is indifferent between the truth-telling (which induces (z_{jk}, p_{jk})) and a slight downward deviation (which induces $(z_{j+1,k}, p_{j+1,k})$), and thus, the worst case could be worse than the truth telling (recall that z_{jk} is monotonic). If F is absolutely continuous, the agents have the threshold types with probability zero, and therefore, we can ignore this subtle issue. If F is atomic, especially at threshold types, we need to be careful about their behaviors.³¹

Proof. See the appendix. □

Therefore, a lower bound of the optimal welfare guarantee is given by the supremum of the welfare guarantee of monotonic LDIC (or MLDIC) mechanisms.

Observe that a dominant-strategy mechanism must satisfy the LDIC conditions and have a monotone trading rule,³² and thus, the class of MLDIC mechanisms is larger than the class of dominant-strategy mechanisms. However, a dominant-strategy mechanism needs to satisfy the upward incentive compatibility constraints as well (i.e., the truth-telling weakly dominates pretending to be more-eager types), and therefore, the welfare guarantee of the optimal dominant-strategy mechanism could be (potentially) strictly lower than the lower bound of the optimal welfare guarantee.

5 When a posted-price mechanism is optimal

In this section, we use the upper bound to show that, for a class of distributions $F(c, v)$, no mechanism can improve upon the optimal posted-price

³¹For example, under a posted-price mechanism with price p , the type $c = p$ or $v = p$ has two admissible messages: accept or reject p , and in the worst-case scenario, these types reject the price.

³²See Hagerty and Rogerson (1987).

mechanism. Throughout this section, we assume that $F(c, v)$ is absolutely continuous with pdf $f(c, v)$.

Theorem 3. Suppose that $w(c, v) \equiv (v - c)f(c, v)$ is strictly decreasing in c , strictly increasing in v , and continuous in (c, v) , for any c, v where $c < v$. Then no mechanism guarantees expected welfare strictly higher than the welfare guarantee of the posted-price mechanism with price p^* , where p^* is given by $w(0, p^*) = w(p^*, 1)$.

Notice that the expected welfare is given by $\int_{c,v} z(c, v)w(c, v) dvdc$, which is the weighted average of $z(c, v)$, where the weight is $w(c, v)$ for each state (c, v) . The monotonicity on the weight function means that more-eager types have higher weights, and in this sense, more-eager types are “more important” in view of the welfare. This condition is satisfied under independent uniform distributions, and any distribution such that $f(c, v)$ is non-increasing in c and non-decreasing in v .

Theorem 1 implies that the upper bound of the optimal welfare guarantee is given by the supremum of the truth-telling expected welfare of LDIC revelation mechanisms, or more specifically,

$$\begin{aligned} \sup_{(c_j)_{j=1}^{J+1}, (v_k)_{k=0}^K, (z_{jk}, p_{jk})_{j,k}} \quad & \sum_{j=1}^J \sum_{k=1}^K \int_{c_{j-1}}^{c_j} \int_{v_k}^{v_{k+1}} (v - c) z_{jk} dF(c, v) \\ \text{sub.to} \quad & (p_{jk} - c_j) z_{jk} \geq (p_{j+1,k} - c_j) z_{j+1,k}, \quad \forall j, k \\ & (v_k - p_{jk}) z_{jk} \geq (v_k - p_{j,k-1}) z_{j,k-1}, \quad \forall j, k \\ & (p_{J+1,k} - 1) z_{J+1,k} = p_{j0} z_{j0} = 0, \quad \forall j, k \\ & 0 \leq c_1 < \dots < c_J \leq c_{J+1} = 1, \quad 0 = v_0 \leq v_1 < \dots < v_K \leq 1. \end{aligned}$$

In the following, we show that it is guaranteed by a posted-price mechanism. As the first step of the proof, we obtain a relaxed problem so that we do not need to find the threshold types.

Lemma 1. If $(z(c, v), p(c, v))_{c,v \in [0,1]}$ is an outcome function of an LDIC rev-

elation mechanism, then for any c, v ,

$$u_1(c, v) \equiv (p(c, v) - c)z(c, v) \geq \int_c^1 z(t, v) dt,$$

$$u_2(c, v) \equiv (v - p(c, v))z(c, v) \geq \int_0^v z(c, t) dt.$$

See the appendix for the proof of this lemma. As a corollary, we obtain the following.

Corollary 1. If $(z(c, v), p(c, v))_{c, v \in [0, 1]}$ is an outcome function of an LDIC revelation mechanism, then for any c, v ,

$$(v - c)z(c, v) \geq \int_c^1 z(t, v) dt + \int_0^v z(c, t) dt.$$

The left hand side is the sum of the utilities or the information rents for the seller of type c and the buyer of type v . Thus, this expression means that, under an LDIC revelation mechanism, the sum of the information rents must be bounded from below. The constraint $(v - c)z(c, v) \geq \int_c^1 z(t, v) dt + \int_0^v z(c, t) dt$ is called a *surplus constraint at (c, v)* (or in short, $SC(c, v)$). Obviously, if $c > v$, then we must have $z(c, v) = 0$.

A similar constraint is obtained by Hagerty and Rogerson (1987) for a dominant-strategy mechanism, although they obtain the same constraint with equality as implication of dominant-strategy incentive compatibility. The LDIC conditions only bound the information rents from below.

We now prove Theorem 3 by considering the relaxed problem for the upper bound.

$$\begin{aligned} & \sup_{z(\cdot)} \int_{c, v} (v - c)z(c, v) dF(c, v) \\ \text{sub.to} \quad & (v - c)z(c, v) \geq \int_c^1 z(t, v) dt + \int_0^v z(c, t) dt, \quad c \leq v. \end{aligned}$$

To give some intuition, we consider a special case with $f(c, v) = 1$ for $(c, v) \in [0, 1]^2$ (i.e., a bivariate uniform distribution), and hence the theorem yields $p^* = \frac{1}{2}$. See the appendix for the proof of the general case.

Consider an even more relaxed problem, which considers only the surplus constraints $SC(1 - q, q)$ for $q \in [\frac{1}{2}, 1]$:

$$(2q - 1)z(1 - q, q) \geq \int_{1-q}^1 z(t, q) dt + \int_0^q z(1 - q, t) dt.$$

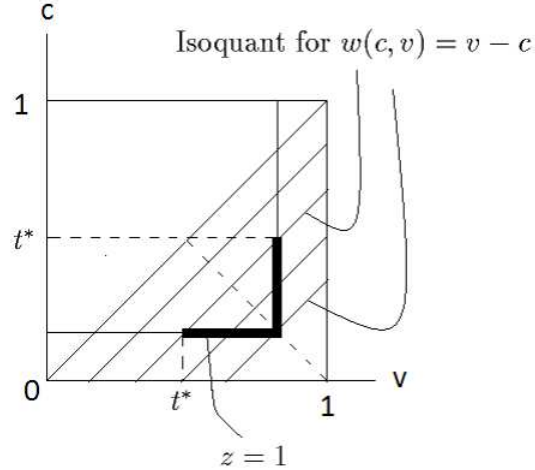
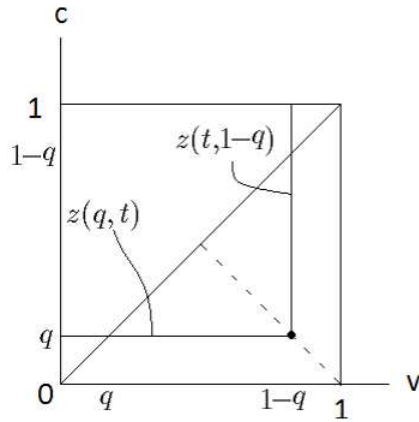
Also, notice that the objective can be decomposed as follows.

$$\begin{aligned} & \int_{c,v} z(c, v)w(c, v) dvdc \\ &= \int_{q=0}^1 \left[\int_{1-q}^1 z(t, q)(q - t) dt + \int_0^q z(1 - q, t)(t - 1 + q) dt \right] dq \\ &= \int_{q=\frac{1}{2}}^1 \left[\int_{1-q}^1 z(t, q)(q - t) dt + \int_0^q z(1 - q, t)(t - 1 + q) dt \right] dq, \end{aligned}$$

where the last equality comes because $z(c, v) = 0$ for $c > v$.

We solve the following decomposed problem separately, and show that the solutions to them also consist of the solution to the original problem. The decomposed problem is, for each $q \in [\frac{1}{2}, 1]$:

$$\begin{aligned} & \max_{z(\cdot)} \int_{1-q}^1 z(t, q)(q - t) dt + \int_0^q z(1 - q, t)(t - 1 + q) dt \\ \text{sub.to} \quad & (2q - 1)z(1 - q, q) \geq \int_{1-q}^1 z(t, q) dt + \int_0^q z(1 - q, t) dt. \end{aligned}$$



In this problem, $z(1-q, q)$ is not constrained from above, and thus, we set $z(1-q, q) = 1$. Given that, the problem is interpreted as a consumer's utility maximization problem with budget constraint (i.e., $z(1-q, t), z(t, q)$ denote the quantities of different goods, the prices are all one, budget is $2q - 1$, and the utility is linear in z). Thus, the solution sets $z(c, v) = 1$ if its weight $w(c, v) = v - c$ is high enough, and sets 0 otherwise. That is, there is w^* such that $z(1-q, t) = 1$ if $q - t \geq w^*$ and $z(t, q) = 1$ if $t - 1 + q \geq w^*$.

Because the weight function is decreasing in c and increasing in v , there is t^* such that $z(1-q, t) = 1$ if and only if $t \geq t^*$, $z(t, q) = 1$ if and only if $t \leq t^*$, and $q - t^* = t^* - 1 + q (= w^*)$, or $t^* = \frac{1}{2}$.

Therefore, at the solution to the decomposed problem for any $q \in [\frac{1}{2}, 1]$, the agents trade if $c < t^* = \frac{1}{2}$ and $v > t^* = \frac{1}{2}$, and do not trade otherwise. A posted-price mechanism with price $p = \frac{1}{2}$ induces this allocation rule. Therefore, the posted-price mechanism with $p = \frac{1}{2}$ is the optimal mechanism.

Remark 3. Because of the symmetry of the distribution implied by a bivariate uniform distribution, in the argument above, the optimal posted-price is given by $p^* = \frac{1}{2}$, and the surplus constraints on the diagonal are binding. For a general distribution, our proof constructs a downward-sloping curve (not necessarily the diagonal) that connects (p^*, p^*) to $(0, 1)$ such that maximizing the decomposed welfare functions subject to the surplus constraints for the points on this curve yields the posted-price mechanism with price p^* .

This result may be interpreted as giving a foundation for the use of dominant-strategy mechanisms as the optimal robust mechanisms for some distributions.

6 Binary types

In this section, we assume that $F(c, v)$ has a finite support (i.e., each $S_i \subseteq [0, 1]$ is a finite set). According to Theorem 1, the upper bound of the welfare guarantee of a finite mechanism is given by an LDIC revelation mechanism where all the threshold types of each agent i are in S_i .

Let $S_1 = \{c_1, \dots, c_J\}$ and $S_2 = \{v_1, \dots, v_K\}$, and let $\mu_{jk} = \Pr((c, v) = (c_j, v_k))$ be the probability mass function for $F(c, v)$. The problem of finding the upper bound of the optimal welfare guarantee is given as follows.

$$\begin{aligned} & \sup_{(z_{jk}, p_{jk})_{j,k}} \sum_{j=1}^J \sum_{k=1}^K (v_k - c_j) z_{jk} \mu_{jk} \\ & \text{sub.to} \quad (p_{jk} - c_j) z_{jk} \geq (p_{j+1,k} - c_j) z_{j+1,k}, \quad \forall j, k, \\ & \quad (v_k - p_{jk}) z_{jk} \geq (v_k - p_{j,k-1}) z_{j,k-1} \quad \forall j, k, \\ & \quad (p_{J+1,k} - 1) z_{J+1,k} = p_{j0} z_{j0} = 0, \quad \forall j, k. \end{aligned}$$

Notice that the problem is linear in each z_{jk}, p_{jk} , and thus, it can be solved by linear programming.

In the previous result, we have shown that the upper bound coincides with the welfare guarantee of the optimal posted-price mechanism. Another question is whether there is a class of environments where we can characterize the optimal mechanism, and it guarantees strictly higher expected welfare than any optimal posted-price mechanism. In the following, we consider binary type spaces: $S_1 = \{c_1, c_2\}$ and $S_2 = \{v_1, v_2\}$, where $0 \leq c_1 < v_1 < c_2 < v_2 \leq 1$. We normalize $c_1 = 0$ and $v_2 = 1$. Let $\mu_{jk} > 0$ for any $j, k = 1, 2$. Then, the problem becomes simpler.

$$\begin{aligned} & \sup_{(z_{jk}, p_{jk})_{j,k}} \sum_{j=1}^2 \sum_{k=1}^2 (v_k - c_j) z_{jk} \mu_{jk} \\ & \text{sub.to} \quad (p_{jk} - c_j) z_{jk} \geq (p_{j+1,k} - c_j) z_{j+1,k}, \quad \forall j, k, \\ & \quad (v_k - p_{jk}) z_{jk} \geq (v_k - p_{j,k-1}) z_{j,k-1} \quad \forall j, k, \\ & \quad (p_{J+1,k} - 1) z_{J+1,k} = p_{j0} z_{j0} = 0, \quad \forall j, k. \end{aligned}$$

Proposition 1. In a solution, we have $(z_{jk}, p_{jk}) = (0, 0)$ for any $j > k$, $p_{11} = v_1$, $p_{22} = c_2$, $z_{12} = 1$, and

- if $\mu_{11}c_2v_1 > \mu_{22}(1-c_2)(1-v_1)$, then $z_{11} = 1$, $z_{22} = \frac{v_1}{c_2} (< 1)$ and $p_{12} = v_1$. In this case, the truth-telling expected welfare is

$$\overline{W} = \mu_{12} + v_1 \left(\mu_{11} + \mu_{22} \frac{1 - c_2}{c_2} \right).$$

- if $\mu_{11}c_2v_1 < \mu_{22}(1 - c_2)(1 - v_1)$, then $z_{22} = 1$, $z_{11} = \frac{1-c_2}{1-v_1} (< 1)$ and $p_{12} = c_2$.³³ In this case, the truth-telling expected welfare is

$$\overline{W} = \mu_{12} + (1 - c_2) \left(\mu_{22} + \mu_{11} \frac{v_1}{1 - v_1} \right).$$

Proof. First, increasing z_{jk} only reduces the expected welfare if $j > k$, and thus, it is set to be zero (and also, $p_{jk} = 0$). For z_{12} , it is not constrained from above, and thus, $z_{12} = 1$ is optimal.

Next, if $p_{11} < v_1$, increasing p_{11} does not reduce the truth-telling expected welfare. On the other hand, p_{11} cannot be higher than v_1 because of the LDIC condition for v_1 . Thus, we set $p_{11} = v_1$. Similarly, we set $p_{22} = c_2$.

Then, the problem becomes as follows.

$$\begin{aligned} \sup_{(z_{jk}, p_{jk})_{j,k=1,2}} \quad & \mu_{12} + v_1 z_{11} \mu_{11} + (1 - c_2) z_{22} \mu_{22} \\ \text{sub.to} \quad & p_{12} \geq c_2 z_{22}, \\ & 1 - p_{12} \geq (1 - v_1) z_{11}. \end{aligned}$$

These two constraints imply $c_2 z_{22} + (1 - v_1) z_{11} \leq 1$. Because the problem is linear in z_{11}, z_{22} , at least one of them is one, and the other is determined in order to bind this new constraint. If we set $z_{11} = 1$, then we obtain $z_{22} = \frac{v_1}{c_2} (< 1)$ and $p_{12} = v_1$. If we set $z_{22} = 1$, then we obtain $z_{11} = \frac{1-c_2}{1-v_1} (< 1)$ and $p_{12} = c_2$. Comparing the two values of the objective, we obtain the solution. $\square$

In the following, we assume $\mu_{11}c_2v_1 > \mu_{22}(1 - c_2)(1 - v_1)$ so that the first case applies, but the same argument works for the second case as well.

Observe that this LDIC revelation mechanism is monotonic. Moreover, it is equivalent to the following two-price mechanism: The seller chooses either of $(\frac{v_1}{c_2}, c_2)$, $(1, v_1)$ or $(0, 0)$ (“opt-out”), and the buyer chooses the highest acceptable price, c_2 , v_1 , or 0 (“opt-out”). Therefore, given Theorem 2, one may hope that this mechanism guarantees $\overline{W}$. However, recall that Theorem

³³If $\mu_{11}c_2v_1 = \mu_{22}(1 - c_2)(1 - v_1)$, then both are the solutions.

2 is proved under an atomless distribution, and it does not directly apply to this situation. Indeed, under this mechanism, the worst-case expected-welfare is lower than $\overline{W}$.³⁴

Nevertheless, we can approximate $\overline{W}$ by a sequence of two-price mechanisms. Specifically, consider a two-price mechanism $\langle M^\varepsilon, g^\varepsilon \rangle$ with $(z_1, p_1) = (\frac{v_1 - \varepsilon}{c_2 + \varepsilon}, c_2 + \varepsilon)$, $(z_2, p_2) = (1, v_1 - \varepsilon)$ with small $\varepsilon > 0$. In this mechanism, pretending to be a less eager type is weakly dominated by the truth-telling for each threshold type. In the limit as $\varepsilon \rightarrow 0$, we obtain $\overline{W}$, and in this sense, we approximately obtain the optimal welfare guarantee.

Notice that the welfare guarantee of the optimal posted-price mechanism, W^{DS} , is strictly lower than $\overline{W}$. The optimal posted-price sets either $p \in (0, \underline{v})$ (which guarantees $\underline{v}\mu_{00} + \mu_{11}$), or $p \in (\bar{c}, 1)$ (which guarantees $\bar{c}\mu_{10} + \mu_{11}$). Neither of them can be strictly higher than $\overline{W}$.³⁵

7 Demand-curve (and supply-curve) mechanisms

As we have discussed, the optimal MLDIC mechanism provides a lower bound of the optimal welfare guarantee, and sometimes, the bound is tight. Thus,

³⁴For example, the buyer of type $v = 1$ has two admissible messages: Reporting $v = 1$ truthfully, or reporting $\underline{v}$ (i.e., pretending to be the low type). Under an atomless distribution, the threshold types' behaviors were negligible.

³⁵The two-price mechanism considered here induces a dominant action for every threshold type. In this sense, we can view this mechanism as a dominant-strategy mechanism (regarding to the binary type spaces). However, if we think of $[0, 1]$ as each agent's "true" type space (although most of them "do not occur"), then the two-price mechanism is not a dominant-strategy mechanism, because, for example, the seller with $c \in (0, v_2)$ has two admissible messages (and neither is dominant). As Hagerty and Rogerson (1987) show, any dominant-strategy mechanism reduces to a posted-price mechanism on this continuum type spaces. Note that as in the example in Section 3, if there is a small probability that each agent's type is different from these two types, the two-price mechanism still guarantees $\overline{W}$ approximately. Thus, one may interpret this section's binary environment as a limit of such "almost binary" environments.

it is important to investigate the optimal MLDIC mechanism in more detail. In particular, under the binary types in the previous section, the optimal mechanism is given by a two-price mechanism, where one of the agents has a dominant strategy.

In this section, we introduce a class of “ K -price” mechanisms as a generalization of two-price mechanisms, and show, under what conditions, the optimal MLDIC mechanism is in that class. Also, we interpret that class of mechanisms as “(fixed) demand-curve” (or “supply-curve”) mechanisms.

Let us introduce demand-curve mechanisms. Recall that, under the “two-price” mechanism in the previous example, the buyer has a dominant strategy, while the seller does not. In that mechanism, the seller can choose one of the allocations, $(z_1, p_1), (z_2, p_2)$, where $z_1 < z_2, p_1 > p_2$, and the buyer can (through a report of his highest acceptable price) only accept or reject each allocation. This idea is generalized to a “ K -price” mechanism: The seller chooses an allocation in $\{(z_1, p_1), \dots, (z_K, p_K)\}$, where $z_k < z_{k+1}, p_k > p_{k+1}$, and the buyer reports the highest acceptable price.

An interpretation of this mechanism is the following: The mechanism designer fixes a decreasing function $z(p)$ at the beginning, which we call a “demand curve”. Given this demand curve, the seller chooses the trading price p_s , the buyer chooses the highest acceptable price p_b , where $p_s, p_b \in \{p_1, \dots, p_K\}$ and the outcome is $(z(p_s), p_s)$ if $p_b \geq p_s$, and $(0, 0)$ otherwise. Specifically, in the previous two-price mechanism, $z(p_1) = z_1$ and $z(p_2) = z_2$.

Definition 4. $\langle M, g \rangle$ is a demand-curve mechanism if, for some K , $M_1 = M_2 = \{0, p_1, \dots, p_K, 1\}$, where $0 < p_1 < \dots < p_K < 1$, and there is a non-increasing function $z(p) : [0, 1] \rightarrow [0, 1]$ such that $g(p_s, p_b) = (z(p_s), p_s)$ if $p_s \leq p_b$, and $g(p_s, p_b) = (0, 0)$ otherwise.³⁶

Under a demand-curve mechanism, the seller’s profit is given by $(p_s - c)z(p_s)$ if p_s is accepted. Notice that $(p_s - c)z(p_s)$ is a monopolist’s profit if we interpret $z(p_s)$ as a demand function. Thus, if the seller believes that the

³⁶Choosing $p_b = 0$ ($p_s = 1$) means the buyer’s (the seller’s) “opting-out”.

“monopoly price” (i.e., the price that maximizes $(p_s - c)z(p_s)$) is acceptable for the buyer, then he chooses that monopoly price to maximize his profit. On the other hand, if he believes that the buyer’s threshold price could be lower, then he has an incentive to choose a lower price. Therefore, in this mechanism, the seller does not have a dominant strategy. The welfare guarantee of the mechanism is obtained when the seller is “optimistic”, i.e., when he believes the monopoly price is acceptable for the buyer. More specifically, the threshold type of the seller between reporting p_{k-1} and p_k is given by $c_k = \frac{p_k z_k - p_{k-1} z_{k-1}}{z_k - z_{k-1}}$, where $z_k = z(p_k)$ and $z_{k-1} = z(p_{k-1})$.

Proposition 2. Suppose that $F(c, v)$ is absolutely continuous with pdf $f(c, v)$, and $w(c, v) = (v - c)f(c, v)$ is non-decreasing in v for $v \geq c$. Then for any MLDIC mechanism with the welfare guarantee W , there is a demand-curve mechanism that guarantees $W' \geq W$.

Proof. See the appendix. □

The optimal demand-curve mechanism is given by the following problem.

$$\max_{K, (z_k, p_k)_{k=1}^K} \sum_{k=1}^K \int_{\frac{p_k z_k - p_{k-1} z_{k-1}}{z_k - z_{k-1}}}^{\frac{p_{k+1} z_{k+1} - p_k z_k}{z_{k+1} - z_k}} (v - c) z_k \, dF(c, v)$$

Similarly, we can define a (fixed) supply-curve mechanism so that an increasing function $z(p)$ is fixed at the beginning, and now the seller chooses the threshold price p_s , the buyer chooses the trading price p_b , where $p_s, p_b \in \{p_1, \dots, p_K\}$ and the outcome is $(z(p_b), p_b)$ if $p_b \geq p_s$, and $(0, 0)$ otherwise. By a similar logic, we obtain the following.

Proposition 3. Suppose that $F(c, v)$ is absolutely continuous with pdf $f(c, v)$, and $w(c, v) = (v - c)f(c, v)$ is non-increasing in c for $c \leq v$. Then for any MLDIC mechanism with the welfare guarantee W , there is a supply-curve mechanism that guarantees $W' \geq W$.

8 Relationship to alternative approaches

8.1 Mechanism with infinitely many messages

The upper bound we have obtained is for the welfare guarantee of the optimal mechanism among all *finite* mechanisms. One may wonder how the results could change if we are allowed to use mechanisms with infinitely many messages.

As shown by Jackson (1992), with infinite message spaces, we can construct a mechanism with an “integer-game” or “tail-chasing” structure, that fully implements essentially any social choice correspondence. In our context, for example, we can construct an unbounded mechanism that guarantees approximately the first-best welfare:

Proposition 4. For any $\varepsilon > 0$, there exists a mechanism with infinitely many messages whose welfare guarantee is at least $(1 - \varepsilon)$ fraction of the first-best welfare.

The idea of the following argument is from Jackson (1992), but we construct a simpler mechanism.

Proof. Consider the following mechanism: The seller reports (c, α) , where $c \in [0, 1]$ and $\alpha \in [0, 1)$. The buyer reports (v, β) , where $v \in [0, 1]$ and $\beta \in [0, 1)$. The outcome function is the following.

- $g(c, v, \alpha, \beta) = (0, 0)$ if $c > v$.
- $g(c, v, \alpha, \beta) = (z(\alpha, \beta), p(c, v, \alpha, \beta))$ if $c \leq v$,

where $z(\alpha, \beta) = 1 - \varepsilon + \frac{\alpha + \beta}{2}\varepsilon$, and

$$p(c, v, \alpha, \beta) = \frac{1}{z(\alpha, \beta)} \left[(1 - \varepsilon) \frac{c + v}{2} + \varepsilon \frac{\alpha c + \beta v}{2} \right].$$

Suppose that the seller reports $(c, 0)$ and the buyer reports $(v, 0)$. Then, the probability of trading is $1 - \varepsilon$, and the price is $\frac{c+v}{2}$. Each agent can, by

increasing α or β , increase the probability of trading slightly. Specifically, the seller can increase z at the price of exactly c , and the buyer can increase z at the price of exactly v .

Therefore, the seller of type c is indifferent between reporting (c, α) and (c, α') for any α, α' . Indeed, any such message is admissible for the seller. Reporting (c', α) where $c' < c$ is dominated by (c, α) , because the seller's utility only goes down by reporting a lower cost.

He may be better off if he reports (c', α) for $c' > c$, because the trading price becomes higher. However, (c', α) is weakly dominated by any (c', α') for $\alpha' > \alpha$. This argument holds for any $\alpha \in [0, 1)$, and therefore, there is no message (c', α) that is admissible for the seller of type c . Therefore, for the seller of type c , any (c, α) is admissible, and no other message is admissible. Similarly, for the buyer of type v , any (v, β) is admissible, and no other message is admissible.

The worst-case scenario of this mechanism is that the agents report their cost and valuation truthfully with $(\alpha, \beta) = (0, 0)$. In that case, the expected welfare is given by $(1 - \varepsilon) \int_{c,v} (v - c) dF(c, v)$. $\square$

This example illustrates a problem of implementation with a mechanism with infinitely many messages. Notice that the seller of cost c would be better off by pretending to be a higher-cost type, but there is no *best* (in the sense of dominance argument) deviation of such kind, and then, we *eliminate* all such deviations. This is essentially the same argument used in mechanisms with an “integer-game” structure.³⁷

Jackson (1992) argues that eliminating an action that is not dominated by any admissible action seems problematic, and offers the class of bounded mechanisms as the mechanisms that do not have such problematic properties.³⁸ In this paper, we have focused on finite mechanisms, because any

³⁷Another (related) problematic feature of such a mechanism is that we cannot define some types' best response action sets. In the mechanism above, given any message of the buyer (v, β) , the set of the best response actions is empty for the seller with cost $c < v$.

³⁸A bounded mechanism is a mechanism such that, if a message m_i of agent i is weakly

finite mechanism is bounded in the sense of Jackson (1992). However, it is an open question if there is any bounded mechanism with infinitely many messages whose welfare guarantee is not approximated by any sequence of finite mechanisms (and so is strictly higher than the upper bound obtained in this paper).

8.2 Alternative solution concepts

In the literature, there are several solution concepts in constructing “robust” mechanisms, other than the admissible strategies.

8.2.1 Mixed-strategy Dominance

So far, we define admissible strategies based on the concept of pure strategy dominance. An alternative approach is to define an admissible strategy as a strategy that is not weakly dominated by any mixed strategies. If we allow mixed strategy dominance, then, given a mechanism, the set of admissible messages becomes (weakly) smaller for any type of any agent. In this sense, we might be able to obtain a more optimistic result. This means that our lower bound is still valid, but our upper bound might no longer be a valid bound. However, it turns out that the same upper bound is valid even if we consider mixed strategy dominance, and therefore, the results in the previous sections remain true.

To be more precise, given a mechanism $\langle M, g \rangle$, for the seller of type c , the set of admissible messages based on mixed strategy dominance is defined

dominated for type θ_i , then there is another message m'_i that is admissible for θ_i and weakly dominates m_i . A mechanism that does not satisfy this property is called an unbounded mechanism.

by

$$\begin{aligned}
M_1^{mixed}(c) &= \{m_1 \in M_1 \mid \nexists \sigma_1 \in \Delta(M_1) \text{ s.t.} \\
&\quad \forall m_2 \in M_2, \sum_{m'_1} \sigma_1(m'_1)(p(m'_1, m_2) - c)z(m'_1, m_2) \geq (p(m) - c)z(m), \\
&\quad \exists m_2 \in M_2, \sum_{m'_1} \sigma_1(m'_1)(p(m'_1, m_2) - c)z(m'_1, m_2) > (p(m) - c)z(m)\},
\end{aligned}$$

where $\Delta(M_1)$ denotes the set of all lotteries on M_1 (and similarly, $M_2^{mixed}(v)$ is defined for the buyer of type v). Let $W^{mixed}(\Gamma)$ denote the welfare guarantee of a mechanism Γ based on mixed strategy dominance. Note that $W^{mixed}(\Gamma)$ is weakly higher than $W(\Gamma)$.

The following proposition shows that $W^{mixed}(\Gamma)$ is approximately upper-bounded by the truth-telling expected welfare of an LDIC revelation mechanism.

Proposition 5. For any mechanism $\Gamma = \langle M, g \rangle$ and any $\delta > 0$, there is an LDIC revelation mechanism $\Gamma^R(\delta)$ with thresholds $\{c_1, \dots, c_J\}$ and $\{v_1, \dots, v_K\}$ such that $c_j \in S_1$ for each j , $v_k \in S_2$ for each k , and

$$\overline{W}(\Gamma^R(\delta)) + \delta \geq W^{mixed}(\Gamma).$$

Note that the LDIC conditions are defined based on pure strategy dominance. Therefore, the same upper bound as in pure strategy dominance is valid even if we consider mixed strategy dominance.

Proof. See the appendix for the proof. $\square$

8.2.2 Iteratively admissible strategies

One may wonder how our result could change if we allow iterative elimination of weakly dominated actions, instead of just one-round of elimination of weakly dominated actions as in our approach.³⁹ This approach is,

³⁹We need to be careful in applying iterative elimination of weakly dominated actions, because the set of actions that survive iterative elimination of weakly dominated actions depends on the specific order of elimination.

roughly speaking, more optimistic than the (one-round) admissible strategies approach. However, actually, it turns out that the results do not change by introducing the iteration.

To be clear, we say that a mechanism $\langle M, g \rangle$ guarantees W in one-round admissible strategies if any admissible strategy profile of the agents induces an expected welfare of at least W . This is the definition of the welfare guarantee in the previous sections.

We define another notion of welfare guarantee, which is based on iteratively admissible actions. Let $\langle M, g \rangle$ be an arbitrary mechanism. Define $M_i^0 = M_i$ for each i . In each round $t = 1, \dots$, given M_i^{t-1} , we define M_i^t as follows.⁴⁰

$$\begin{aligned} M_1^t(c) = & \{m_1 \in M_1^{t-1} \mid \nexists m'_1 \in M_1^{t-1} \text{ s.t.} \\ & \forall m_2 \in M_2^{t-1}, (p(m'_1, m_2) - c)z(m'_1, m_2) \geq (p(m) - c)z(m), \\ & \exists m_2 \in M_2^{t-1}, (p(m'_1, m_2) - c)z(m'_1, m_2) > (p(m) - c)z(m)\}, \end{aligned}$$

and $M_1^t = \bigcup_{c \in [0,1]} M_1^t(c)$. Define M_2^t similarly.

$M_1^t(c)$ is the set of admissible messages for type c of the seller given that the buyer chooses any message in M_2^{t-1} . Then, M_1^t is given as the union of $M_1^t(c)$, that is, any message of the seller that survives the t -th round of elimination is admissible for some type of the seller as long as the buyer chooses a message in M_2^{t-1} . In this sense, this is an incomplete-information version of the iterative elimination. We stop the procedure when no message can be eliminated (and we stop at some finite step because M_i is finite), and define M_i^* as the set of messages that survive this process. Any $m_i \in M_i^*$ is admissible for some type of agent i as long as the opponent chooses a message in M_{-i}^* .

Finally, for each c of the seller, define $M_1^*(c)$ as the set of admissible messages for type c if the buyer chooses a message in M_2^* , and similarly, for

⁴⁰The following definition is based on pure strategy dominance. However, applying the same logic as in the previous subsection, the results remain true even if we consider mixed strategy dominance.

each v of the buyer, define $M_2^*(v)$ as the set of admissible messages for type v if the seller chooses a message in M_1^* . We say that a mechanism $\langle M, g \rangle$ guarantees W in iteratively admissible strategies if

$$W = \int_{c,v} \left[\min_{m \in M^*(c,v)} z(m)(v - c) \right] dF(c, v),$$

where $M^*(c, v) = M_1^*(c) \times M_2^*(v)$ and $g(m) = (z(m), p(m))$.

Notice that, once we obtain M_i^* as the surviving message sets for each agent i , we evaluate a mechanism as if this is the original message space. More precisely, let g^* be the restriction of g on M^* , and consider a new mechanism $\langle M^*, g^* \rangle$. This mechanism guarantees W in one-round admissible strategies, by the definition of W , and at the same times, this guarantees W in iteratively admissible strategies, because any $m_i \in M_i^*$ is admissible for some type of agent i , and thus, the elimination process stops at the first stage.

Proposition 6. Let $\langle M, g \rangle$ be an arbitrary mechanism that guarantees W in iteratively admissible strategies. Then there is $\langle M^*, g^* \rangle$ that guarantees W in one-round admissible strategies, and also in iteratively admissible strategies.

On the other hand, the approach of the one-round admissible strategies may be more optimistic than that of iteratively admissible strategies.⁴¹ This can happen if the optimal mechanism in one-round admissible strategies has some messages never used by any type of an agent, yet this message is used by the other agents to eliminate some undesirable messages. In this case, our lower bound may not be valid in the approach of iteratively admissible strategies.⁴² However, notice that our lower bound is obtained by an MLDIC revelation mechanism, and under an MLDIC revelation mechanism, any message of an agent is admissible at least for some type of the agent. Therefore, it guarantees W in one-round admissible strategies if and only if it

⁴¹See Dasgupta, Hammond, and Maskin (1979).

⁴²More precisely, (poentially) there may be no mechanism that attains the lower bound in iteratively admissible strategies.

guarantees W in iteratively admissible strategies. Specifically, the two-price mechanisms guarantee the same expected welfare even if they are evaluated in iteratively admissible strategies, and they continue to be optimal under the same corresponding assumptions.

8.2.3 Strict dominance

One of the more pessimistic approaches is to allow each agent to take any strategy that is not strictly dominated. Our upper bound is still valid for this approach, but under this behavioral assumption, we obtain much more pessimistic result. Namely, we cannot essentially avoid “no-trading equilibria” by any mechanism that satisfies (ex post) individual rationality and budget balance, because an agent’s message corresponding to “opt-out” makes the opponent indifferent among all messages. Therefore, in the previous sections, we need to eliminate weakly dominated actions to make a sensible argument. By the same reason, we cannot avoid no-trading equilibria in iterative elimination of strictly dominated strategies.

8.2.4 Epistemic approach

A recent literature on game theory and mechanism design discusses which actions an agent may take under several behavioral assumptions that are based on common knowledge of rationality among the agents, but without common prior assumption. For example, Battigalli (1999) and Battigalli and Siniscalchi (2003) study “incomplete information rationalizability”.⁴³ An action is said to be incomplete information rationalizable for an agent of a given type, if it survives the iterative deletion process of dominated actions for the agent of that type (in the sense of mixed strategy dominance).

Under complete information, a rationalizable action has its epistemic foundation called an a posteriori equilibrium.⁴⁴ Similarly, an action that

⁴³This is a special class of “ Δ -rationalizability” in Battigalli and Siniscalchi (2003). See also Bergemann and Morris (2007).

⁴⁴An a posteriori equilibrium is a subjective correlated equilibrium with a refinement on

is incomplete information rationalizable has its epistemic foundation, which is an incomplete-information analogue of an a posteriori equilibrium, and called an “interim equilibrium”.⁴⁵

Notice that the concept of the incomplete information rationalizability is based on iterative deletion of strictly dominated actions. Therefore, if we directly apply this approach, we must have “no-trading” equilibria again. In this sense, the standard epistemic approach is more pessimistic than our approach.⁴⁶ However, as we see in the next section, the two approaches coincide with expected (but not exact) budget balance requirement.

8.3 Without (exact) budget balance

In this section, we modify the model to allow non-budget-balanced allocations. The allocation is now a triple (z, s, t) , where $z \in [0, 1]$ is the probability of trading, the same as before, but s denotes the monetary transfer to the seller, and t denotes the monetary transfer to the buyer. Originally, they satisfy $s = -t = zp$ for the trading price p .

A simple story would be that the mechanism designer can be a residual claimant in order to balance the net transfers. In this case, the expected (not exact or ex post) budget balance may be the only necessary requirement, if

probability zero events. See Aumann (1987) for the definition of an a posteriori equilibrium. Brandenburger and Dekel (1987) show that an action is rationalizable for an agent if and only if there is an a posteriori equilibrium under which the agent takes that action.

⁴⁵More rigorously, an interim equilibrium is defined given a specific description of the agents’ possible belief structures, or “a type space”, where a type of an agent here describes not only his payoff type but also a belief (and higher-order beliefs) on the opponents’ types. See Bergemann and Morris (2005b), Bergemann and Morris (2007) and the “Bayesian equilibrium model” of Battigalli and Siniscalchi (2003). Bergemann and Morris (2005b), Bergemann and Morris (2005a), and Bergemann and Morris (2009) study the relationship between implementation in all interim equilibria and implementation in ex post equilibria (which reduces to dominant-strategy implementation in private values).

⁴⁶For iterative admissibility, the corresponding epistemic concept would be an “undominated” interim equilibrium, that is, an interim equilibrium consisting only of admissible strategies.

she is not budget constrained. Then the optimal LDIC mechanism is given by the following problem. Let $\lambda > 0$ denote the “shadow price” for the expected budget balance constraint.⁴⁷

$$\begin{aligned} \max_{z,s,t} \quad & \int_{c,v} (v - c)z(c, v) dF(c, v) - \lambda \int_{c,v} [s(c, v) + t(c, v)] dF(c, v) \\ \text{sub.to} \quad & u_1(c, v)(\equiv s(c, v) - cz(c, v)) = \int_c^1 z(t, v) dt, \quad c \leq v \\ & u_2(c, v)(\equiv vz(c, v) + t(c, v)) = \int_0^v z(c, t) dt, \quad c \leq v. \end{aligned}$$

The constraints are the “threshold-free” expressions of LDIC constraints (see Lemma 1), where the left hand sides are the agents’ utilities, or their information rents at state (c, v) . However, the difference from the exact budget balance case is that all of these LDIC conditions with equality.⁴⁸ The binding LDIC condition for agent i at state (c, v) is called $ICFOC_i(c, v)$. Replacing $s(c, v), t(c, v)$ by the $ICFOC_i(c, v)$, the objective function becomes the following.

$$\begin{aligned} & \int_{c,v} (v - c)z(c, v) dF(c, v) - \lambda \int_{c,v} \left[\int_c^1 z(t, v) + cz(c, v) + \int_0^v z(c, t) dt - vz(c, v) \right] dF(c, v) \\ = & \int_{c,v} \left[((1 + \lambda)v - \lambda \frac{1 - F(v|c)}{f(v|c)}) - ((1 + \lambda)c + \frac{F(c|v)}{f(c|v)}) \right] z(c, v) dF(c, v), \end{aligned}$$

where the equality comes by applying integration by parts. Suppose that the monotone hazard rate conditions are satisfied for the conditional distributions: $\frac{1 - F(v|c)}{f(v|c)}$ is non-increasing in v for any $v \geq c$, and $\frac{F(c|v)}{f(c|v)}$ is non-decreasing

⁴⁷An alternative situation is that the mechanism designer is a government who is concerned not only about the agents’ welfare, but also the “tax payers”. Then, she may desire to maximize the weighed sum of the expected welfare of the agents and the tax payers, as in Laffont and Tirole (1993). In this case, λ represents the shadow price of the transfer from the tax payers to the agents. This becomes equivalent to a revenue maximization problem, if $\frac{1}{\lambda} \rightarrow 0$.

⁴⁸Otherwise, we can decrease $s(c, v)$ or $t(c, v)$ by a small amount without violating any other constraints, and can increase the objective. With the exact budget balance, this logic does not apply, because any decrease in the transfer to one of the agents implies an increase in the transfer to the other agent.

in c for any $c \leq v$, then we obtain a monotone trading rule as the solution to this expected welfare maximization problem. Moreover, this monotone trading rule is implemented by a dominant-strategy mechanism.⁴⁹

This conclusion is related to a result obtained by Chung and Ely (2007). They study, among others, the mechanisms that maximize the worst-case expected revenue, and show that such a mechanism is a dominant-strategy mechanism, if $F(c, v)$ satisfies affiliation and monotone hazard rate condition. Note that the “worst-case” in their definition is different from this paper. They consider a situation that the agents play a Bayesian Nash equilibrium under a mechanism, but the mechanism designer does not know their beliefs (as to the opponents’ preferences) correctly, and therefore, the “worst case” means the agents’ beliefs under which the expected revenue in a given mechanism is minimized.

9 Conclusion

In this paper, we have established an upper and a lower bound of the welfare guarantee achievable in any finite mechanism. Under some distributions of the agents’ private information, the upper bound implies no improvement over posted-price mechanisms of Hagerty and Rogerson (1987). On the other hand, with binary types, a two-price mechanism guarantees strictly higher expected welfare than any posted-price mechanism, and it is indeed (approximately) optimal among all finite mechanisms.

It is interesting to study whether the results could extend to other economic problems such as double auction, public-good provision, two-sided matching, and so on.

⁴⁹It is because, under the monotonic ICFOC mechanism, each threshold type is completely indifferent between the truth-telling and the slight downward deviation, and therefore, any type below the threshold type has no incentive to deviate upward. Thus, the mechanism must also satisfy upward incentive compatibility, and hence, the mechanism is a dominant-strategy mechanism. Furthermore, by perturbing the mechanism, each agent can have a strict dominant strategy.

A Proof of Theorem 1

We first show the following lemma.

Lemma 2. For any mechanism $\Gamma = \langle M, g \rangle$, there is a set of the seller's types $\{c_1, \dots, c_J\}$ such that $0 \leq c_1 < \dots < c_J \leq 1$ and $c_j \in S_1$ for each j , a set of the buyer's types $\{v_1, \dots, v_K\}$ such that $0 \leq v_1 < \dots < v_K \leq 1$ and $v_k \in S_2$ for each k , a set of messages in M_1 , $\{a_1, \dots, a_J\}$, and a set of messages in M_2 , $\{b_0, \dots, b_K\}$, satisfying the following. Let $c_{J+1} = 1$, $v_0 = 0$, and let $C_1 = [0, c_1]$, $C_j = (c_{j-1}, c_j]$ for $j \geq 2$, $V_k = [v_k, v_{k+1})$ for $k \leq K-1$, and $V_K = [v_K, 1]$. Also, let $(z_{jk}, p_{jk}) = g(a_j, b_k)$ for each j, k .

(i) For each j , if a_j is not admissible for type $c \in C_j$, then $c \notin S_1$, and for each k , if b_k is not admissible for type $v \in V_k$, then $v \notin S_2$. (ii) For each j and k ,

$$\begin{aligned} (p_{jk} - c_j)z_{jk} &\geq (p_{j+1,k} - c_j)z_{j+1,k}, \\ (v_k - p_{jk})z_{jk} &\geq (v_k - p_{j,k-1})z_{j,k-1}, \\ (p_{J+1,k} - 1)z_{J+1,k} &= p_{j0}z_{j0} = 0. \end{aligned}$$

Proof. We only construct the buyer's threshold types and messages. A similar logic applies for the seller as well.

For each $b \in M_2$, let $V(b)$ be the set of types for whom b is admissible.⁵⁰ Because it is defined by finite number of inequalities, it consists of finitely many subintervals of $[0, 1]$.⁵¹ Also, $\bigcup_{b \in M_2} V(b) = [0, 1]$.

The following lemma is useful.

⁵⁰We assume any two message $b, b' \in M_2$ are not equivalent in the sense that there is some $a \in M_1$ such that $g(a, b) \neq g(a, b')$ without loss of generality.

⁵¹More precisely,

$$\begin{aligned} V(b) = & \bigcap_{b' \neq b} \left[\left(\bigcup_{a \in M_1} \{v | (v - p(a, b))z(a, b) > (v - p(a, b'))z(a, b')\} \right) \right. \\ & \left. \cup \left(\bigcap_{a \in M_1} \{v | (v - p(a, b))z(a, b) = (v - p(a, b'))z(a, b')\} \right) \right]. \end{aligned}$$

The second set is either empty or singleton, and the first set consists of finite number of intervals.

Lemma 3. For some $b \in M_2$ and v , suppose that $v \in V(b)$, and there is a sequence $v(n) \downarrow v$ such that $v(n) \notin V(b)$ for any n . Then there exist b' , n' , and $\bar{\varepsilon} > 0$ such that for any $\varepsilon \in (0, \bar{\varepsilon})$, $v + \varepsilon \in V(b')$, and for any $n > n'$, b' weakly dominates b for $v(n)$. Moreover, v is indifferent between b and b' for any $a \in M_1$ (and hence $v \in V(b')$).

Proof. (of Lemma 3) Suppose that there is $b(n) \in M_2$ that weakly dominates b for $v(n)$ but not for $v(n')$, where $n' > n$. Then, there is $a \in M_1$ such that $v(n')$ prefers $b(n)$ to b as a response but $v(n')$ does not. The single crossing property implies that for any $n'' > n'$, $b(n)$ does not weakly dominate b for type $v(n'')$. Because there are finitely many messages, there must be some b' such that b' weakly dominates b for type $v(n)$ any sufficiently large n . Also, because $V(\tilde{b})$ consists of finitely many subintervals for each $\tilde{b} \in M_2$, there is $\bar{\varepsilon} > 0$ such that at least one such b' satisfies $v + \varepsilon \in V(b')$ for $\varepsilon \in (0, \bar{\varepsilon})$.

Then, b' weakly dominates b for any type $v(n) \downarrow v$ means that $(v - p(a, b))z(a, b) \leq (v - p(a, b'))z(a, b')$ for any $a \in M_1$. However, since $v \in V(b)$, all the inequality must hold with equality; otherwise b' weakly dominates b for type v . Thus, v is indifferent between b and b' for any $a \in M_1$, and hence $v \in V(b')$. $\square$

Now we construct the threshold types and messages. We begin the following procedure with $t = 0$, $v_0 = 0$, and any admissible message of v_0 , $b_0 \in M_2$.

(Step $t = 0, 1, \dots$) Note that $v_t \in S_2$ and $v_t \in V(b_t)$.

If $\{v \geq v_t | v \in S_2, v \notin V(b_t)\} = \emptyset$, then we finish the procedure. Let $v' = \inf\{v \geq v_t | v \in S_2, v \notin V(b_t)\}$. Since S_2 is closed, $v' \in S_2$. There are three cases. (i) $v' > v_t$ and $v' \notin V(b_t)$, (ii) $v' > v_t$ and $v' \in V(b_t)$, or (iii) $v' = v_t$.

For the first case, let $b_{t+1} \in M_2$ be an admissible message of v' that weakly dominates b_t . Let $v_{t+1} = v'$, and we proceed to Step $t + 1$.

For the second case, there is a sequence $v(n) \downarrow v'$ such that $v(n) \notin V(b_t)$. Then, there must be b' and n' such that, for any $n > n'$, $v(n) \in V(b')$ and b'

weakly dominates b_t for $v(n)$. Also, v' is indifferent between b_t and b' . Let $v_{t+1} = v'$, $b_{t+1} = b'$, and we proceed to Step $t + 1$.

For the third case, there is a sequence of $v(n) \downarrow v_t$ such that $v(n) \notin V(b_t)$. Then, there must be b' and n' such that, for any $n > n'$, $v(n) \in V(b')$ and b' weakly dominates b_t for $v(n)$. Also, v_t is indifferent between b_t and b' . Let $v_t = v'$, $b_t = b'$, and we repeat Step t (and then, either of the first two cases applies because of the lemma above).

Notice that this procedure finishes within finite steps. If not, we obtain a message b and a sequence of types $v^1 < v^2 \dots$ such that b is admissible for v^j , j is odd, and not admissible for v^k , k is even. This contradicts that $V(b)$ consists of finitely many subintervals.

By construction, each threshold is in S_2 , b_k is admissible for $v \in V_k$ as long as $v \in S_2$. $\square$

Let Γ^R be a revelation mechanism with threshold types $\{c_1, \dots, c_J\}$ and $\{v_1, \dots, v_K\}$ and the outcome function $(z(c, v), p(c, v)) = (z_{jk}, p_{jk})$ for each $c \in C_j$ and $v \in V_k$. Then, by construction, it is LDIC. Also,

$$\begin{aligned} \overline{W}(\Gamma^R) &= \int_{c,v} (v - c)z(c, v) dF(c, v) \\ &= \sum_{j=1}^{J+1} \sum_{k=0}^K \int_{(c,v) \in C_j \times V_k} (v - c)z_{jk} dF(c, v). \end{aligned}$$

The proof completes if we can show that $\overline{W}(\Gamma^R) \geq W(\Gamma)$. Reporting $c \in (c_{j-1}, c_j]$ in this revelation mechanism corresponds to reporting a_j in Γ , and similarly, reporting $v \in [v_k, v_{k+1})$ corresponds to reporting b_k . Note that, by construction, reporting a_j is admissible for any $c \in C_j \cap S_1$. Similarly,

reporting b_k is admissible for any $v \in V_k \cap S_2$. Therefore,

$$\begin{aligned}\overline{W}(\Gamma^R) &= \sum_{j=1}^{J+1} \sum_{k=0}^K \int_{(c,v) \in C_j \times V_k} (v - c) z_{jk} dF(c, v) \\ &= \sum_{j=1}^{J+1} \sum_{k=0}^K \int_{(c,v) \in (C_j \cap S_1) \times (V_k \cap S_2)} (v - c) z_{jk} dF(c, v) \\ &\geq W(\Gamma).\end{aligned}$$

B Proof of Theorem 2

First, we see the threshold types' incentives. Under the LDIC mechanism, if the LDIC condition for the seller of type c_j ,

$$(p_{jk} - c_j) z_{jk} \geq (p_{j+1,k} - c_j) z_{j+1,k},$$

holds with equality for all k , then type c_j has at least two admissible messages: Reporting c_j truthfully (which attains (z_{jk}, p_{jk}) for each k), and reporting $c \in (c_j, c_{j+1})$ (which attains $(z_{j+1,k}, p_{j+1,k})$ for each k). Similarly, if the LDIC condition for the buyer of type v_k ,

$$(v_k - p_{jk}) z_{jk} \geq (v_k - p_{j,k-1}) z_{j,k-1},$$

holds with equality for all k , then type v_k has at least two admissible messages: Reporting v_k truthfully, and reporting $v \in (v_{k-1}, v_k)$ (which attains $(z_{j+1,k}, p_{j+1,k})$ for each k). However, their behaviors do not affect the expected welfare if F is not atomic at the threshold types. Therefore, it suffices to show that, for any other types, the truth-telling weakly dominates any downward deviation.

Consider the seller's type $c \in (c_{j-1}, c_j)$. If it is admissible for this type to report $c' \in (c_j, c_{j+1})$, an "adjacent" downward deviation, then we must have

$$(p_{jk} - c) z_{jk} \leq (p_{j+1,k} - c) z_{j+1,k},$$

for some k . However, because

$$(p_{jk} - c_j)z_{jk} \geq (p_{j+1,k} - c_j)z_{j+1,k},$$

the single-crossing property and monotonicity in z_{jk} implies $z_{jk} = z_{j+1,k}$ and $p_{jk} = p_{j+1,k}$. Since this is true for any k such that $(p_{jk} - c)z_{jk} \leq (p_{j+1,k} - c)z_{j+1,k}$, we conclude that, if it is admissible for type c to report $c' \in (c_j, c_{j+1})$, then these two messages are equivalent, and hence, no adjacent downward deviation essentially occurs by non-threshold types.

Also, if for some $c \in (c_{j-1}, c_j)$ the truth-telling weakly dominates reporting $c' \in (c_j, c_{j+1})$, then for any $c'' < c$, the truth-telling weakly dominates reporting c' as well. Therefore, the mechanism is globally downward incentive compatible (for any non-threshold types).

C Proof of Lemma 1

We first show the inequalities for the threshold types.

For the seller of type c_j , for each k , the LDIC condition implies

$$c_j(z_{jk} - z_{j+1,k}) \leq p_{jk}z_{jk} - p_{j+1,k}z_{j+1,k}.$$

Thus, summing both sides for $j' = j, \dots, J$,

$$\sum_{j'=j}^J c_{j'}(z_{j'k} - z_{j'+1,k}) \leq \sum_{j'=j}^J (p_{j'k}z_{j'k} - p_{j'+1,k}z_{j'+1,k}),$$

Because $(p_{J+1,k} - 1)z_{J+1,k} = 0$, we have

$$c_j z_{jk} + \sum_{j'=j+1}^{J+1} (c_{j'} - c_{j'-1}) z_{j',k} \leq p_{jk} z_{jk},$$

where $c_{J+1} = 1$, and thus,

$$u_1(c_j, v_k) \equiv (p_{jk} - c_j)z_{jk} \geq \sum_{j'=j+1}^{J+1} (c_{j'} - c_{j'-1}).$$

Because $(c_{j'} - c_{j'-1})z_{j',k} = \int_{c_{j'-1}}^{c_{j'}} z(t, v) dt$, we obtain

$$u_1(c_j, v_k) \geq \int_{c_j}^1 z(t, v_k) dt.$$

Now, consider $c \in (c_{j-1}, c_j]$ and $v \in [v_k, v_{k+1})$. because $z(c, v) = z_{jk}$ and $p(c, v) = p_{jk}$, we have

$$u_1(c, v) = u_1(c_j, v_k) + (c_j - c)z_{jk} \geq \int_c^1 z(t, v_k) dt = \int_c^1 z(t, v) dt.$$

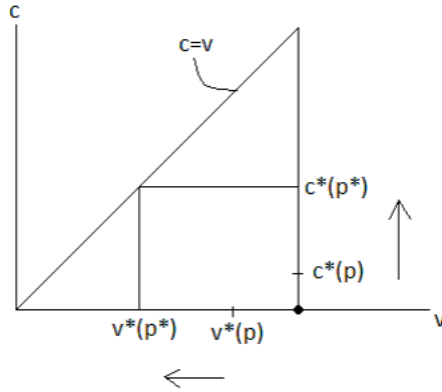
Similarly, for the buyer of any type v , we obtain

$$u_2(c, v) \geq \int_0^v z(c, t) dt.$$

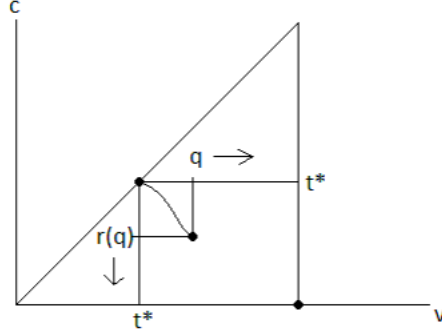
D Proof of Theorem 3

For each $p \in [0, f(0, 1)]$, let $c^*(p)$ be c such that $p = (1 - c)f(c, 1)$. For each p , $c^*(p)$ uniquely exists (because of the assumption above), and $c^*(0) = 1$, $c^*(f(0, 1)) = 0$, and it is continuous and decreasing in p . Similarly, let $v^*(p)$ be v such that $p = vf(0, v)$. Then $v^*(p)$ uniquely exists for each p , $v^*(0) = 0$, $v^*(f(0, 1)) = 1$, and it is continuous and increasing in p .

Thus, there is p^* such that $c^*(p^*) = v^*(p^*)$. Define $t^* = c^*(p^*)$. Note that $p^* = w(t^*, 1) = w(0, t^*)$.



For each $q \in [t^*, 1]$, define $r(q)$ so that $w(t^*, q) = w(r(q), t^*)$. For each q , $r(q)$ uniquely exists (because of the assumption above), $r(t^*) = t^*$, $r(1) = 0$ (because $w(t^*, 1) = w(0, t^*)$), and $r(q)$ is continuous and decreasing in q . These imply $r(q) \leq t^* \leq q$ for any q .



Recall that our (relaxed) problem is

$$\begin{aligned} \sup_{z(\cdot)} \quad & \int_{c,v} (v - c) z(c, v) dF(c, v) \\ \text{sub.to} \quad & (v - c) z(c, v) \geq \int_c^1 z(\theta, v) d\theta + \int_0^v z(c, \theta) d\theta, \quad c \leq v. \end{aligned}$$

In the following, we ignore all the constraints except for the surplus constraint at $(r(q), q)$ for each $q \in [t^*, 1]$:

$$SC(r(q), q) : (q - r(q)) z(r(q), q) \geq \int_{r(q)}^1 z(\theta, q) d\theta + \int_0^q z(r(q), \theta) d\theta.$$

Also, notice that the objective can be decomposed as follows.

$$\begin{aligned} \int_{c,v} z(c, v) w(c, v) dv dc &= \int_{q=0}^1 \left[\int_{r(q)}^1 z(\theta, q) w(\theta, q) d\theta + \int_0^q z(r(q), \theta) w(r(q), \theta) d\theta \right] dq \\ &= \int_{q=t^*}^1 \left[\int_{r(q)}^1 z(\theta, q) w(\theta, q) d\theta + \int_0^q z(r(q), \theta) w(r(q), \theta) d\theta \right] dq, \end{aligned}$$

where the last equality comes because $z(c, v) = 0$ for $c > v$.

Now, consider the following decomposed problem. For each $q \in [t^*, 1]$,

$$\begin{aligned} \max_{z(\cdot)} \quad & \int_{r(q)}^1 z(\theta, q) w(\theta, q) d\theta + \int_0^q z(r(q), \theta) w(r(q), \theta) d\theta \\ \text{sub.to} \quad & (q - r(q)) z(r(q), q) \geq \int_{r(q)}^1 z(\theta, q) d\theta + \int_0^q z(r(q), \theta) d\theta. \end{aligned}$$

A solution sets $z(r(q), q) = 1$. Given this, the problem is a sort of a utility maximization problem with budget constraint. (each z is quantity, prices are all one, and the utility is linear in z). Because of the assumption above, we find $\bar{c}, \underline{v}$ such that (i) $z(r(q), v) = 1$ iff $v \geq \underline{v}$, (ii) $z(c, q) = 1$ iff $c \leq \bar{c}$, and (iii) $w(r(q), \underline{v}) = w(\bar{c}, q)$. By the constraint, $q - r(q) = (q - \underline{v}) + (\bar{c} - r(q))$, or $\underline{v} = \bar{c}$. Let $\hat{t} = \underline{v}$.

At $\hat{t}$, we have $w(r(q), \hat{t}) = w(\hat{t}, q)$. Because $w(r(q), t)$ is increasing in t and $w(t, q)$ is decreasing in t , $\hat{t}$ is uniquely determined. This implies that $\hat{t} = t^*$, because $w(r(q), t^*) = w(t^*, q)$.

Let $W^*(q)$ denote the value of the problem above. Then the welfare guarantee is no greater than $\int_{t^*}^1 W^*(q) dq$. Thus, if we can find a mechanism that guarantees $\int_{t^*}^1 W^*(q) dq$, then it is optimal. However, this level of expected welfare is guaranteed by a posted-price mechanism with price t^* , because under the mechanism, we obtain $z(c, v) = 1$ iff $c \leq t^*$ and $v \geq t^*$.

E Proof of Proposition 2

Let $\Gamma^R = (z(c, v), p(c, v))$ denote an arbitrary MLDIC mechanism, with threshold types $\{c_1, \dots, c_J\}$ and $\{v_1, \dots, v_K\}$. That is, $(z(c, v), p(c, v)) = (z_{jk}, p_{jk})$ for each $c \in C_j$ and $v \in V_k$.

Fix any j in the following. Suppose $c \in (c_{j-1}, c_j)$. Let k be the smallest integer such that $z_{jk} > 0$, and suppose $v \in (v_k, v_{k+2})$. Originally, the buyer with $v \in (v_k, v_{k+1})$ is assigned (z_{jk}, p_{jk}) , and $v \in (v_{k+1}, v_{k+2})$ is assigned $(z_{j,k+1}, p_{j,k+1})$ under the truth-telling.

Imagine the following hypothetical situation. The seller report's his type truthfully. The buyer can, instead of the truthful report, choose either $(0, 0)$

or $(z_{j,k+1}, p_{j,k+1})$ (i.e., suppose that (z_{jk}, p_{jk}) is unavailable). Then, any type $v < p_{j,k+1}$ chooses $(0, 0)$, and any type $v > p_{j,k+1}$ chooses $(z_{j,k+1}, p_{j,k+1})$.

Conditional on $c \in (c_{j-1}, c_j)$ and $v \in (v_k, v_{k+2})$, the expected welfare is

$$\frac{1}{A} \int_{c_{j-1}}^{c_j} \int_{p_{j,k+1}}^{v_{k+2}} z_{j,k+1}(v - c) f(c, v) \, dv dc,$$

where $A = \Pr((c, v) \in (c_{j-1}, c_j) \times (v_k, v_{k+2}))$.

Originally, the expected welfare conditional on $c \in (c_{j-1}, c_j)$ and $v \in (v_k, v_{k+2})$ was at most

$$\begin{aligned} & \frac{1}{A} \int_{c_{j-1}}^{c_j} \left(\int_{v_k}^{v_{k+1}} z_{jk}(v - c) f(c, v) dv + \int_{v_{k+1}}^{v_{k+2}} z_{j,k+1}(v - c) f(c, v) dv \right) dc \\ & \leq \frac{1}{A} \int_{c_{j-1}}^{c_j} \left(\int_{p_{jk}}^{\frac{p_{j,k+1}z_{j,k+1} - p_{jk}z_{jk}}{z_{j,k+1} - z_{jk}}} z_{jk}(v - c) f(c, v) dv \right. \\ & \quad \left. + \int_{\frac{p_{j,k+1}z_{j,k+1} - p_{jk}z_{jk}}{z_{j,k+1} - z_{jk}}}^{v_{k+2}} z_{j,k+1}(v - c) f(c, v) dv \right) dc, \end{aligned}$$

because of the LDIC conditions. Therefore, subtracting the second from the first,

$$\begin{aligned} & \frac{1}{A} \int_{c_{j-1}}^{c_j} \left(- \int_{p_{jk}}^{p_{j,k+1}} z_{jk}(v - c) f(c, v) dv \right. \\ & \quad \left. + \int_{p_{j,k+1}}^{\frac{p_{j,k+1}z_{j,k+1} - p_{jk}z_{jk}}{z_{j,k+1} - z_{jk}}} (z_{j,k+1} - z_{jk})(v - c) f(c, v) dv \right) dc \\ & \geq \frac{1}{A} \int_{c_{j-1}}^{c_j} \left(-(p_{j,k+1} - p_{jk}) z_{jk}(p_{j,k+1} - c) f(c, p_{j,k+1}) dv \right. \\ & \quad \left. + \left(\frac{p_{j,k+1}z_{j,k+1} - p_{jk}z_{jk}}{z_{j,k+1} - z_{jk}} - p_{j,k+1} \right) (z_{j,k+1} - z_{jk})(p_{j,k+1} - c) f(c, p_{j,k+1}) dv \right) dc \\ & \geq 0. \end{aligned}$$

The first inequality comes because $(v - c)f(c, v)$ is increasing in v . This means that, if $(v - c)f(c, v)$ is increasing in v , then we can improve by eliminating an allocation that is used by lower-valuation buyers. By the elimination, some lower-valuation buyers ($v \in (v_k, p_{j,k+1})$ in the case above)

lose the opportunity of trading, which decreases the expected welfare. On the other hand, it makes some higher-valuation buyers ($v \in (p_{j,k+1}, v_{k+1})$ in the case above) switch from a lower-probability allocation to a higher-probability allocation (from (z_{jk}, p_{jk}) to $(z_{j,k+1}, p_{j,k+1})$), which increases the expected welfare. The inequality above shows that the net effect is positive if $(v - c)f(c, v)$ is increasing in v .

Applying this argument consecutively, we eliminate all the allocations except for (z_{jK}, p_{jK}) (and $(0, 0)$), without decreasing the expected welfare conditional on $c \in (c_{j-1}, c_j)$. Also, this argument applies to any j . Therefore, we finally obtain the following (worst-case) allocation rule: At state (c, v) where $c \in (c_{j-1}, c_j)$ and $v \in (p_{jK}, 1)$, the allocation is (z_{jK}, p_{jK}) , and $(0, 0)$ otherwise. Let W be the expected welfare induced by this allocation rule. Then, a demand curve mechanism such that the seller chooses one of $(z_{1K}, p_{1K}), \dots, (z_{JK}, p_{JK})$ (or no trading) and the buyer reports a threshold price guarantees the expected welfare no smaller than W , because for each j , we have

$$c_j(z_{jK} - z_{j+1,K}) \leq p_{jK}z_{jK} - p_{j+1,K}z_{j+1,K},$$

or $\frac{p_{jK}z_{jK} - p_{j+1,K}z_{j+1,K}}{z_{jK} - z_{j+1,K}} \geq c_j$ by the LDIC condition of the original mechanism. The left hand side is the threshold type under the demand-curve mechanism, and because this is true for any j , the welfare guarantee cannot be lower than W .

F Proof of Proposition 5

For each v , define $M_2^*(v)$ as the set of messages that is a strict best response for some $\sigma_1 \in \Delta(M_1)$.

$$\begin{aligned} M_2^*(v) = & \{b \in M_2 \mid \exists \sigma_1 \in \Delta(M_1), \forall \sigma_2 \in \Delta(M_2), \\ & \sum_a \sigma_1(a)[u_2(g(a, b), v)] > \sum_{a, b'} \sigma_1(a)\sigma_2(b')[u_2(g(a, b'), v)]\}. \end{aligned}$$

Let $V^*(b) \subseteq [0, 1]$ be the set of the buyer's types v such that $b \in M_2^*(v)$.

Lemma 4. If $v \in V^*(b)$, then there exists $\varepsilon > 0$ such that for any $v' \in (v - \varepsilon, v + \varepsilon)$, $v' \in V(b)$.

Proof. If $v \in V^*(b)$, then there exists $\sigma_1 \in \Delta(M_1)$ such that for any $\sigma_2 \in \Delta(M_2)$,

$$\sum_a \sigma_1(a) [u_2(g(a, b), v)] > \sum_{a, b'} \sigma_1(a) \sigma_2(b') [u_2(g(a, b'), v)].$$

This inequality holds for $v' \in (v - \varepsilon, v + \varepsilon)$ if ε is small enough. $\square$

This implies that $V^*(b)$ is an open set, and so it consists of at most countably many open subintervals of $[0, 1]$.⁵² Then, for any $\delta(b) > 0$, there is a subset of $V^*(b)$, say $V^-(b)$ such that $\Pr(v \in V^-(b)) \leq \delta(b)$ and $V^*(b) \setminus V^-(b)$ consists of finitely many subintervals of $[0, 1]$.

Let $\delta = \sum_b \delta(b)$, and $V^\delta = [0, 1] \setminus [\bigcup_b V^-(b)]$. Note that V^δ is closed.

Construct $C^*(a)$ and $C^-(a)$ for each $a \in M_1$, and C^δ similarly. Define $S_1^\delta = S_1 \cap C^\delta$ and $S_2^\delta = S_2 \cap V^\delta$. In the following, we treat S_i^δ as if this is the support of F (i.e., we “ignore” types outside S_i^δ).

Now, we can apply the same procedure as in Lemma 2 to obtain a finite sequence of threshold types and messages for each agent, except that each message in this sequence may be given by a lottery over messages. Given a mechanism $\Gamma = \langle M, g \rangle$, let $\alpha \in \Delta(M_1)$ and $\beta \in \Delta(M_2)$. Then, let $g(\alpha, \beta)$ be the allocation that is equivalent to the lotteries over allocations induced by (α, β) , i.e.,

$$g(\alpha, \beta) = \sum_{(a, b) \in M} \alpha(a) \beta(b) g(a, b).$$

Lemma 5. For any mechanism $\Gamma = \langle M, g \rangle$ and δ , there is a set of the seller’s types $\{c_1, \dots, c_J\}$ such that $0 \leq c_1 < \dots < c_J \leq 1$ and $c_j \in S_1^\delta$ for each j , a set of the buyer’s types $\{v_1, \dots, v_K\}$ such that $0 = v_0 \leq v_1 < \dots < v_K \leq 1$ and $v_k \in S_2^\delta$ for each k , a set of lotteries on M_1 , $\{\alpha_1, \dots, \alpha_J\}$, and a set of

⁵²This is because each interval must contain distinct rational number, and there are at most countably many rational numbers.

messages in M_2 , $\{\beta_0, \dots, \beta_K\}$, satisfying the following. Let $c_{J+1} = 1$, $v_0 = 0$, and let $C_1 = [0, c_1]$, $C_j = (c_{j-1}, c_j]$ for $j \geq 2$, $V_k = [v_k, v_{k+1})$ for $k \leq K-1$, and $V_K = [v_K, 1]$. Also, let $(z_{jk}, p_{jk}) = g(\alpha_j, \beta_k)$ for each j, k .

(i) For each j , if there is $a \in M_1$ in the support of α_j that is not admissible for type $c \in C_j$, then $c \notin S_1^\delta$, and for each k , if there is $b \in M_2$ in the support of β_k that is not admissible for type $v \in V_k$, then $v \notin S_2^\delta$. (ii) For each j and k ,

$$\begin{aligned}(p_{jk} - c_j)z_{jk} &\geq (p_{j+1,k} - c_j)z_{j+1,k}, \\ (v_k - p_{jk})z_{jk} &\geq (v_k - p_{j,k-1})z_{j,k-1}, \\ (p_{J+1,k} - 1)z_{J+1,k} &= p_{j0}z_{j0} = 0.\end{aligned}$$

To prove this lemma, we first show the following lemma that corresponds to Lemma 3.

Lemma 6. For some $b \in M_2$ and $v \in V^\delta$, suppose that $v \in V(b)$, but there is a sequence $v(n) \downarrow v$ such that $v(n) \notin V(b)$ for any n . Then there exists $\beta \in \Delta(M_2)$, n' and $\bar{\varepsilon} > 0$ such that for any $\varepsilon \in [0, \bar{\varepsilon})$ and any b' in the support of β , $v + \varepsilon \in V(b')$, and for any $n > n'$, β weakly dominates b for $v(n)$. Moreover, v is indifferent between b and β for any $a \in M_1$.

Proof. (of Lemma 6)

For each n , there is $\beta(n) \in \Delta(M_2)$ that weakly dominates b for each n . Without loss of generality, we assume that each $b'(n)$ in the support of $\beta(n)$ is in $M_2^*(v(n))$. Basically the same logic in the proof of Lemma 3 applies, but there, we could assume that the same message weakly dominates b for a large enough n because of the finiteness of (pure) messages. However, in this case, $\beta(n)$ can be different for each n . Therefore, we need a different logic to find an appropriate β .

Because $\Delta(M_2)$ is compact, there is a converging subsequence of $\beta(n)$. Let β be its limit. Then, the support of β is a subset of the support of $\beta(n)$ for a large enough n , and also, by definition of V^δ , there is $\bar{\varepsilon} > 0$ such that each member in the support of β is admissible for $v + \varepsilon$ for any $\varepsilon \in [0, \bar{\varepsilon})$. $\square$

Proof. (of Lemma 5)

We apply the same procedure as in the proof of Lemma 2. The only difference is that, for each step t , b_{t+1} is replaced by a mixed action β_{t+1} . We omit the rest of the procedure. $\square$

Define a revelation mechanism as in the proof of Theorem 1, but only on $C^\delta \times V^\delta$: $(z(c, v), p(c, v)) = (z_{jk}, p_{jk})$ for $(c, v) \in C_j \times V_k$. The mechanism satisfies LDIC on $C^\delta \times V^\delta$.

The upper bound for the welfare guarantee of the original mechanism is given as follows:

$$W(\Gamma) \leq \int_{(c,v) \in C^\delta \times V^\delta} z(c, v)(v - c) dF(c, v) + \delta,$$

because of the lemma, and $\int_{(c,v) \notin C^\delta \times V^\delta} (v - c) dF(c, v) \leq \delta$.

Let $\Gamma^R(\delta)$ denote the LDIC revelation mechanism given δ , and let $\overline{W}(\Gamma^R(\delta))$ be its truth-telling expected welfare. As $\delta \rightarrow 0$, the limit of the right hand side coincides with the limit of $\overline{W}(\Gamma^R(\delta))$, and therefore, the LDIC revelation mechanism provides an upper bound in the limit.

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